LECTURE NOTES

Simulations and algorithmic applications of Markov chains

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Lecture notes

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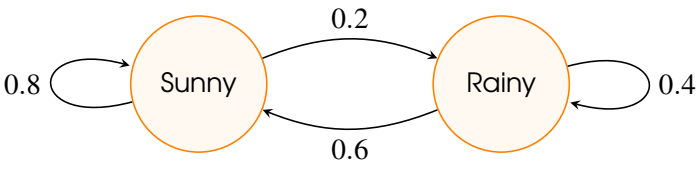
Chapter 1

A first look at Markov chains

1.1 A first example: tomorrow's weather

Suppose we describe each day as either sunny or rainy. We want to generate a sequence of days at random, with some persistence in the weather: a sunny day should make another sunny day more likely. Here is a toy model:

Today's weather	Tomorrow sunny	Tomorrow rainy
Sunny	0.8	0.2
Rainy	0.6	0.4



For example, after a sunny day we toss a coin with success probability 0.8. On success tomorrow is sunny; otherwise it is rainy. After a rainy day we use success probability 0.6. We use fresh independent random choices each day.

Write X_k for the weather on day k. Once we know today's weather, we know which rule to use for tomorrow. Yesterday's weather and the weather a week ago supply no further information about tomorrow *in this model*. This is the idea of the *Markov property*.

States and the Markov property. In general we study a sequence $X_0, X_1, X_2, \ldots$ whose values lie in a finite set of possible states,

$$\mathbb{E} = \{\mathbf{e}_1, \ldots, \mathbf{e}_M\}.$$

For the weather example, $\mathbf{e}_1$ means sunny and $\mathbf{e}_2$ means rainy, so $M = 2$. A state could also be an ordered deck, a colouring of a graph, or a route through several cities. We call $\mathbb{E}$ the *state space*.

We say that the sequence is a Markov chain when, conditional on the present state, the next state does not depend on the earlier history. In probability notation, for any history with positive probability,

$$\mathbb{P}(X_{k+1} = \mathbf{e}_j \mid X_k = \mathbf{e}_i, X_{k-1}, \ldots, X_0) = \mathbb{P}(X_{k+1} = \mathbf{e}_j \mid X_k = \mathbf{e}_i).$$

The next state is still random. The condition says which information we need in order to determine its probabilities. In this course we usually use the same transition rule at each step, as in the weather table above.

1.1.1 Does the starting weather matter after many days?

Start with a sunny day: $X_0 = \mathbf{e}_1$. The probability of sunshine tomorrow is 0.8. For the day after tomorrow, follow the branches of the tree:

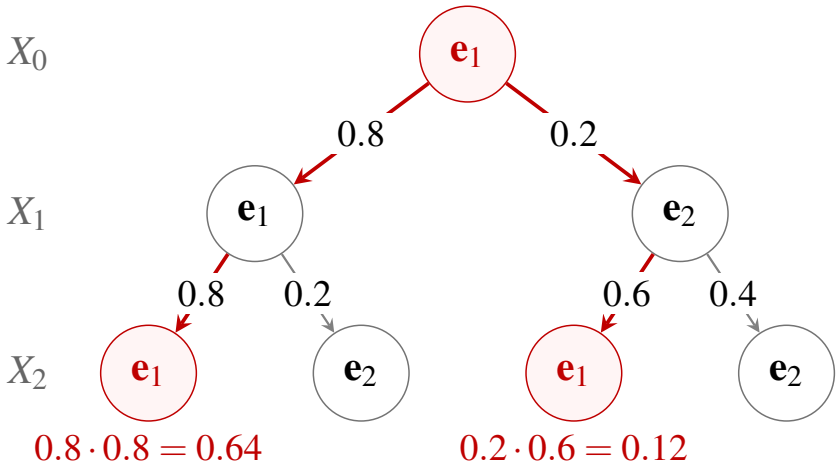


Figure 1.1: Two steps of the weather model, starting from sunshine. The red paths end at $X_2 = \mathbf{e}_1$; $\mathbf{e}_1$ means sunny and $\mathbf{e}_2$ means rainy.

There are two ways to end in sunshine: sunny–sunny–sunny and sunny–rainy–sunny. Multiply the probabilities along each path, then add these two disjoint possibilities:

$$\mathbb{P}(X_2 = \mathbf{e}_1 \mid X_0 = \mathbf{e}_1) = 0.8 \cdot 0.8 + 0.2 \cdot 0.6 = 0.64 + 0.12 = 0.76.$$

The same transition table applies on both days. The probabilities of the next branches depend only on the current weather, which is the Markov property.

If we instead start with a rainy day, the corresponding probabilities are 0.6 and $0.6 \cdot 0.8 + 0.4 \cdot 0.6 = 0.72$. Continuing the calculation gives

Probability of sunshine	Day 0	Day 1	Day 2	Day 3	Day 4
Start sunny	1	0.8	0.76	0.752	0.7504
Start rainy	0	0.6	0.72	0.744	0.7488

Both rows approach 0.75. After many days the model gives almost the same probability of sunshine, regardless of the weather on day zero.

There is a short way to see the pattern. Let $r_k = \mathbb{P}(X_k = \mathbf{e}_1)$. Conditioning on day k gives

$$r_{k+1} = 0.8 r_k + 0.6(1 - r_k) = 0.6 + 0.2 r_k.$$

Since $0.75 = 0.6 + 0.2 \cdot 0.75$, we obtain

$$r_{k+1} - 0.75 = 0.2(r_k - 0.75).$$

Each day reduces the difference from 0.75 by a factor of five. This answers two questions: the limiting probability is 0.75, and we can calculate how fast the initial weather loses its influence.

1.1.2 What is a stationary distribution?

Suppose we choose the weather on day zero at random: sunny with probability $3/4$ and rainy with probability $1/4$. The table gives

$$\mathbb{P}(X_1 = \mathbf{e}_1) = \frac{3}{4} \cdot 0.8 + \frac{1}{4} \cdot 0.6 = \frac{3}{4}.$$

The same calculation applies on each following day. Thus the probabilities of the two states remain

$$\pi(\mathbf{e}_1) = \frac{3}{4}, \qquad \pi(\mathbf{e}_2) = \frac{1}{4}.$$

We call these probabilities a *stationary distribution*. The weather keeps changing; its distribution stays the same.

For many finite Markov chains we will prove that, under suitable assumptions,

$$\mathbb{P}(X_k = \mathbf{e}_i \mid X_0 = \mathbf{e}_j) \longrightarrow \pi(\mathbf{e}_i) \qquad \text{as } k \to \infty, \tag{1.1}$$

with the same limit for every starting state $\mathbf{e}_j$. At a finite time the probabilities need only be *close* to their limits. They equal $\pi(\mathbf{e}_i)$ at every time if we start with the stationary distribution. We will also meet chains for which convergence fails, for example one that alternates between two states at every step.

The weather model is small enough to calculate by hand. For a deck of cards or a large graph, even listing the states can be impossible. The course will ask how to construct useful chains and how to estimate their convergence without listing all their states.

1.2 How many times should we shuffle a deck?

Under suitable assumptions, the distribution of X_k approaches the stationary distribution $\boldsymbol{\pi}$, as in (1.1). For each state $\mathbf{e}_i$, its probability approaches $\pi(\mathbf{e}_i)$, regardless of the starting state. Usually this is a limiting statement: after a finite number of steps the probabilities are not exactly equal to their stationary values.

This leaves a natural question: *how close are we after k steps?* More often we ask the question the other way round: *how large should k be to make the distribution close enough to $\boldsymbol{\pi}$?* These are questions about the *rate of convergence to stationarity*. We will give "close enough" a precise meaning later. For shuffling a deck, the target is a uniform order: each permutation should have probability $1/n!$. Thus convergence alone does not tell us how many times to shuffle. The following example gives us a way to study that question.

1.2.1 Top-to-random: a stopping rule that gives a uniform deck

Take a deck of n distinct cards in a specified initial order. One *top-to-random* move consists of taking the top card and inserting it into one of the n positions, chosen with equal probabilities. Position one is the top, so this choice can leave the deck unchanged.

The state X_k is the whole ordered deck after k moves. It is a Markov chain: the current order and a fresh choice of insertion position determine the next order. Here $\mathbb{E}$ consists of all $n!$ permutations. The probabilities we want are $\pi(\mathbf{e}) = 1/n!$ for each deck order $\mathbf{e}$.

To see how this shuffle creates a uniform order, let b denote the card at the bottom of the *initial* deck. We follow this same card throughout the shuffle; b is its label, not its changing position.

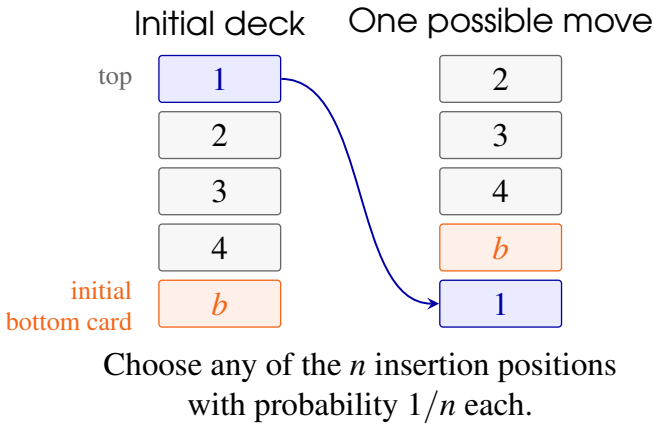


Figure 1.2: A top-to-random move: take the top card and insert it into a uniformly chosen position. Here it enters below the initially bottom card b.

1. Wait until the first card is inserted below b.
2. Continue until there are two cards below b, then three, and so on.
3. When all the other cards lie below b, the card b is at the top. Make one more top-to-random move, inserting b into a uniformly chosen position, and stop. Call the total number of moves T.

Why does this give a uniform deck? With one card below b there is only one possible order in that region. When a second card enters, it can go before or after the first, with equal probabilities. When a third enters, it can go into any of three slots, again with equal probabilities. Continuing this argument shows that, when b reaches the top, all $(n-1)!$ orders of the cards below it have equal probabilities. Inserting b into one of n positions then makes each of the $n!$ full deck orders equally likely.

The waiting times do not favour one of those orders: conditional on a card entering the region below b, its insertion slot is uniform, while the chance of entering depends only on the number of cards already there. Thus the deck is uniform even if we are told the exact stopping time T. This stronger property makes T a *strong stationary time*.

How long do we wait? If there are j cards below b, there are $j+1$ insertion positions below it. The chance that the next move adds a card to that region is $(j+1)/n$. The mean waiting time for that event is therefore $n/(j+1)$. Summing the stages, including the final move of b, gives

$$\mathbb{E}T = n + \frac{n}{2} + \frac{n}{3} + \cdots + \frac{n}{n} = nH_n \sim n\log n, \tag{1.2}$$

where $H_n = 1 + 1/2 + \cdots + 1/n$ and log is the natural logarithm. The first card to enter below b takes n moves on average; the next takes $n/2$, and later stages become faster. This is the source of the $n\log n$ number, rather than an unexplained rule of thumb.

A mean waiting time alone does not tell us how likely a given run is to have finished. To decide on a fixed number L of shuffles, we need a bound on $\mathbb{P}(T > L)$. Later we will prove that this tail probability also bounds the error in the distribution of the deck after L moves.

1.2.2 Riffle shuffles and the story of seven

The shuffle just described moves one card at a time. A familiar hand shuffle cuts the deck into two packets and riffles them together, preserving the order within each packet. We need a different model for that operation.



In one standard model, each of the n cards contributes an independent fair coin toss to the cut size, so the first packet has a binomial number of cards. To interleave the packets, choose the next card from the left packet with probability equal to its fraction of the remaining cards, and otherwise choose from the right packet.

How many repetitions make a 52-card deck close to a uniform permutation? Bayer and Diaconis computed the following values for this model [BD92]:

Number of shuffles	5	6	7	8	9
Distance from uniform	0.924	0.614	0.334	0.167	0.085

We will define this *total variation distance* later. For now, read a value ε as follows: for any event concerning the deck, its probability differs from the probability under a uniform deck by at most ε. Zero means the two distributions agree.

The table explains the famous "seven shuffles" story: a few extra shuffles reduce the error substantially. Seven shuffles do not give an exactly uniform deck. The required number depends on what shuffle we use and what error we are prepared to accept.

Exercise 1.1 Can the top card be uniform among the n labels while the order of the rest of the deck remains highly predictable? Give an example. This helps distinguish randomising one observation from randomising the whole state. ∎

For example, start with $(1,2,\dots,n)$, choose one card uniformly and move it to the top, leaving the others in their original order. The top label is uniform, but, once we know it, the rest of the deck is determined. Only n deck orders are possible, rather than all $n!$.

1.3 A secret permutation: why mixing matters in cryptography

1.3.1 The insertion positions form a shared key

Consider a toy cipher for the symbols $1,\dots,n$. Both the sender and the receiver know the initial deck $(1,2,\dots,n)$ and the top-to-random procedure. Their secret shared key is a sequence of insertion positions

$$\boldsymbol{\theta} = (\theta_1,\dots,\theta_L), \qquad \theta_k \in \{1,\dots,n\}.$$

They apply L top-to-random moves, using position θ_k at move k. Afterwards their decks agree and have some order $\boldsymbol{\sigma} = (\sigma_1,\dots,\sigma_n)$.

Use this order as a substitution: encrypt symbol i as $\sigma(i) = \sigma_i$, the label in position i. The receiver knows the same permutation and decrypts by applying its inverse. Once the key is fixed, both computations are deterministic. For the probability calculations below, assume that the key entries are independent and uniform on $\{1,\dots,n\}$.

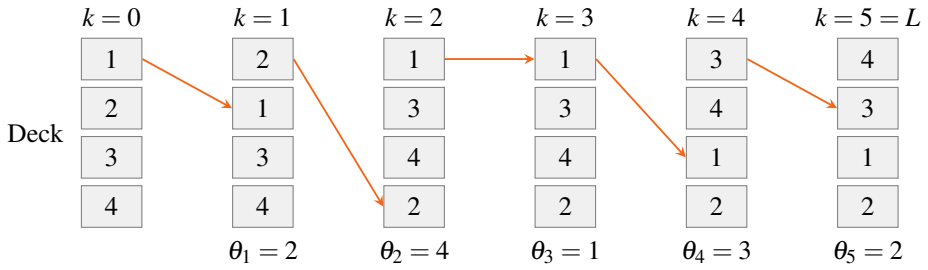


Figure 1.3: Top-to-random with the shared key $\boldsymbol{\theta} = (2,4,1,3,2)$. Each column gives the deck after one more move. The arrow shows where the previous top card is inserted; θ_k specifies that position. Here the key length is $L = 5$.

For example, this key turns $(1,2,3,4)$ into $(4,3,1,2)$. The resulting substitution sends 1 to 4, 2 to 3, 3 to 1 and 4 to 2. The insertion positions in Figure 1.3 explain exactly how the sender and receiver obtain that permutation.

1.3.2 An attacker knows that the key has length one

Suppose the attacker knows the initial order and the procedure, but not the key. If $n = 4$ and $L = 1$, there are only four possible keys. They produce

Key θ_1	Deck after the move	Encryption of symbol 4
1	$(1,2,3,4)$	4
2	$(2,1,3,4)$	4
3	$(2,3,1,4)$	4
4	$(2,3,4,1)$	1

Since the four keys are equally likely,

$$\mathbb{P}\{\sigma(4) = 4\} = \frac{3}{4}. \tag{1.3}$$

For an ideal uniform random permutation, this probability would be $1/4$: any of the four labels is equally likely to occupy the fourth position. An attacker who does not know the key can therefore predict this particular substitution with a substantial advantage after one shuffle.

We want to choose L large enough that the random permutation generated by the key is close to uniform. For instance, if its total variation distance from uniform is at most ε, then

$$\left|\mathbb{P}\{\sigma(4) = 4\} - \frac{1}{4}\right| \leq \varepsilon.$$

The same bound applies to other events concerning the permutation. Thus a question about key length leads directly to a question about the *speed of convergence* of a Markov chain. The stopping rule from the previous section will help us choose a sufficient L. This is a teaching example about randomisation. Reusing one substitution throughout a message preserves repeated-symbol patterns and symbol frequencies, even if we chose the permutation uniformly. This weakness is separate from the bias caused by a short shuffling key; it applies regardless of the key length.

1.3.3 RC4: a historical example with a permutation of 256 symbols

RC4 gives a real reason to study related questions. It maintains a permutation of the byte values $0,\ldots,255$, together with two indices. A procedure based on a shared key initializes the permutation.

Further swaps generate a sequence of bytes, which the sender and receiver use to encrypt and decrypt a message.

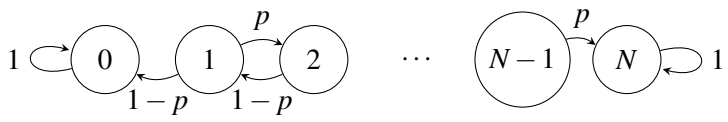
As with our toy cipher, a fixed key gives a deterministic computation. We study probabilities when the key or other inputs are random. The questions include whether some outputs are more likely than others and whether an attacker can exploit those differences. RC4 has exploitable output biases; RFC 7465 prohibited its use in TLS in 2015 [Pop15]. In this course we use the story to motivate the analysis of random permutations and their observable statistics.

1.4 Will the gambler reach the target before going bankrupt?

A gambler starts with i units of capital. At each play the capital increases by one with probability p and decreases by one with probability $1-p$. The gambler stops at capital zero or at a target N. If X_k is the capital after k plays, then

$$\mathbb{E} = \{0, 1, \ldots, N\}.$$

To put this in our general notation, we could label the states $\mathbf{e}_1 = 0, \mathbf{e}_2 = 1, \ldots, \mathbf{e}_{N+1} = N$; the capital values are more convenient here. The current capital determines the next move, so this is another Markov chain. After stopping, we keep X_k at the boundary reached.



The first question is the chance of reaching N before zero. In a fair game, starting with i units gives success probability i/N. For example, with initial capital 3 and target 10, the probability of success is $3/10$. Why is it that simple? What changes if $p \neq 1/2$? How long can play last?

We can begin by conditioning on the first play: the success probability from i is a weighted average of the success probabilities from $i-1$ and $i+1$. This gives equations we can solve. We will also see how martingales and *duality* give other approaches. A duality relates two chains so that a question about one can be answered by studying the other. In some applications, an absorption time of a second chain will even describe convergence of the first.

1.5 Sampling when the list of possibilities is too large

Suppose we want to draw a state $\mathbf{e} \in \mathbb{E}$ with prescribed probabilities $\pi(\mathbf{e})$. If the set is small, we can list the probabilities, split the interval $[0,1]$ into intervals of those lengths, and use a uniform random number to select a state. What if we cannot make the list?

Example 1.1 — A random proper colouring of a graph. Take a graph and a palette of q colours. A colouring is *proper* if the endpoints of each edge have different colours. Let $\mathbb{E}$ be the set of proper colourings and let $Z = |\mathbb{E}|$. We want each colouring to have probability

$$\pi(\mathbf{e}) = 1/Z, \qquad \mathbf{e} \in \mathbb{E}.$$

Even counting the colourings can be difficult, let alone listing them to choose one uniformly.

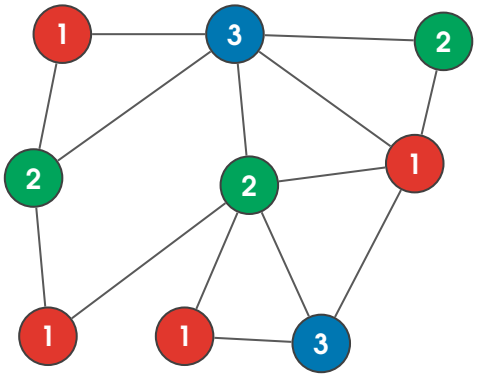


Figure 1.4: A proper colouring: the endpoints of each edge have different colours.

Here is an algorithm we can run without knowing Z:

1. Start with a proper colouring.
2. Choose a vertex uniformly at random.
3. Choose its new colour uniformly from the colours not currently used by its neighbours, including its present colour.
4. Repeat.

Each update keeps the colouring proper. The current colouring determines the possible updates, so the sequence of colourings is a Markov chain. The uniform distribution is stationary for this rule: if we begin with a uniform colouring, one update leaves it uniform.

To turn this into a sampling method, we need more. Can the updates reach all proper colourings? How many updates bring the distribution close to uniform? With too few colours, a local update can get trapped. With sufficiently many colours, we can prove that the method works and estimate its running time. The description of the algorithm is short; proving that its output has the required distribution is the main mathematical task.

This is the idea of *Markov chain Monte Carlo*, or MCMC: design simple random moves whose stationary probabilities are the probabilities we want, then run the chain long enough to obtain an approximate sample. For a non-uniform target we may know only weights $w(\mathbf{e})$, with

$$\pi(\mathbf{e}) = \frac{w(\mathbf{e})}{Z}, \qquad Z = \sum_{\mathbf{e}\in\mathbb{E}} w(\mathbf{e}).$$

The sum can be too large to compute. We will learn update rules, including Metropolis and Gibbs updates, which use local information or ratios of weights and do not require evaluating Z.

1.6 Approximate counting: how many colourings are there?

Sampling asks for one random colouring. Counting asks a different question: *how many proper colourings does the graph have?* For a graph G with v vertices and q available colours, let

$$Z = |\mathbb{E}|, \qquad \mathbb{E} = \{\text{proper } q\text{-colourings of } G\}.$$

There are q^v assignments to check before we even exclude those that are not proper. Can we estimate Z without checking them all?

1.6.1 Why even an approximation can be difficult

For arbitrary graphs, a fast exact counting algorithm would answer whether $Z = 0$. With three colours, deciding whether a proper colouring exists is NP-complete [GJS76]. Thus, assuming

P $\neq$ NP, there is no deterministic polynomial-time algorithm that computes this count for every graph.

Replacing the exact count by an approximation does not automatically solve the difficulty. Suppose an algorithm always returned a number $\widehat{Z}$ with, say, relative error at most 10%:

$$0.9Z \le \widehat{Z} \le 1.1Z.$$

It would still distinguish $Z = 0$ from $Z > 0$, and hence decide three-colourability. This rules out such an always-correct deterministic polynomial-time algorithm on arbitrary graphs under the same assumption. Here the word *relative* matters: a very large additive error could give an easy but uninformative answer.

1.6.2 An answer with a prescribed probability of success

A randomized approximation algorithm offers a different guarantee. We specify a relative tolerance $\varepsilon \in (0,1)$ and a failure probability $\delta \in (0,1)$, and ask for a random estimate satisfying

$$\mathbb{P}\big((1-\varepsilon)Z \le \widehat{Z} \le (1+\varepsilon)Z\big) \ge 1-\delta. \tag{1.4}$$

For example, $\varepsilon = 0.05$ and $\delta = 0.01$ mean an estimate within 5% of the true count with probability at least 99%. The probability is over the random choices made by the algorithm; the graph and its count are fixed. We generally cannot tell from one run whether that run succeeded. Repeating independent runs and taking their median can increase confidence.

If the running time is polynomial in the input size, $1/\varepsilon$, and $\log(1/\delta)$, we call this a *fully polynomial randomized approximation scheme*, or FPRAS. Such schemes exist for useful classes of problems, but randomization does not remove every computational obstruction. We will need assumptions on the graph and the number of colours. We are not claiming that every graph-counting problem becomes easy, or that every useful approximation must be randomized.

1.6.3 Counting via almost uniform sampling

For graph colourings, approximate counting can be reduced to almost uniform sampling, together with estimates of probabilities and some extra bookkeeping [JVV86]. We will develop that construction later. For now, the important connection is that an algorithm for drawing colourings can also help estimate how many colourings exist.

MCMC provides the samples. Its convergence rate tells us how long to run the chain to get sufficiently close to uniform. The final count also depends on random fluctuations in the samples, so we must choose the sample size and mixing accuracy to obtain the requested error and confidence.

There are concrete positive results. If the graph's maximum degree is Δ and $q \ge 2\Delta + 1$, Jerrum's single-vertex recolouring chain mixes in polynomial time and yields an FPRAS for the number of colourings [Jer95]. This sufficient condition is not an optimal threshold. We will return to the details after learning to bound convergence rates.

1.7 Finding a short route by random search

A salesperson must visit each of n cities once and return to the starting city. The *travelling salesperson problem* (TSP) asks for the shortest tour. A state is a visiting order $\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_n)$, a permutation of the cities. With a distance matrix $\mathbf{M}$, its total length is

$$\ell(\boldsymbol{\sigma}) = \sum_{i=1}^{n-1} \mathbf{M}(\sigma_i, \sigma_{i+1}) + \mathbf{M}(\sigma_n, \sigma_1).$$

Our task is to find $\operatorname{argmin}_{\boldsymbol{\sigma} \in \mathbb{E}} \ell(\boldsymbol{\sigma})$. There are $n!$ visiting orders. We can find an exact optimum by enumerating them, but that becomes impractical as n grows.

1.7.1 Turn a short tour into a likely state

For any $\beta > 0$, we have the same optimizing permutations in

$$\begin{aligned}\operatorname*{argmin}_{\boldsymbol{\sigma}\in\mathbb{E}} \ell(\boldsymbol{\sigma}) &= \operatorname*{argmax}_{\boldsymbol{\sigma}\in\mathbb{E}} [-\ell(\boldsymbol{\sigma})] = \operatorname*{argmax}_{\boldsymbol{\sigma}\in\mathbb{E}} e^{-\ell(\boldsymbol{\sigma})} \\ &= \operatorname*{argmax}_{\boldsymbol{\sigma}\in\mathbb{E}} e^{-\beta\ell(\boldsymbol{\sigma})} = \operatorname*{argmax}_{\boldsymbol{\sigma}\in\mathbb{E}} \frac{e^{-\beta\ell(\boldsymbol{\sigma})}}{Z_\beta}.\end{aligned}$$

Here $Z_\beta = \sum_{\boldsymbol{\sigma}\in\mathbb{E}} e^{-\beta\ell(\boldsymbol{\sigma})}$ is a positive normalizing constant independent of the chosen tour. The last expression is a probability distribution:

$$\pi_\beta(\boldsymbol{\sigma}) = \frac{e^{-\beta\ell(\boldsymbol{\sigma})}}{Z_\beta}. \tag{1.5}$$

The shortest tours are precisely its most probable states. Maximizing this function directly can still require expensive search. Instead, suppose we can *sample* from it.

If $\boldsymbol{\sigma}'$ is longer than $\boldsymbol{\sigma}$ by $a > 0$, then $\pi_\beta(\boldsymbol{\sigma}')/\pi_\beta(\boldsymbol{\sigma}) = e^{-\beta a}$: shorter tours receive more weight. For a finite state space, increasing β concentrates the law near the shortest tours. If $\ell_{\min} = \min_{\boldsymbol{\sigma}\in\mathbb{E}} \ell(\boldsymbol{\sigma})$, then a sample $\boldsymbol{\sigma} \sim \boldsymbol{\pi}_\beta$ satisfies, for every $\varepsilon > 0$,

$$\mathbb{P}\big(\ell(\boldsymbol{\sigma}) \le \ell_{\min} + \varepsilon\big) \longrightarrow 1 \qquad (\beta \to \infty).$$

Thus accurate sampling at a sufficiently large β gives a high probability of a good solution. Obtaining that sample is the computational task; a large β can make mixing slow.

1.7.2 A simple procedure for TSP

Start with one tour. Choose two distinct positions uniformly and swap their cities, obtaining a proposed tour $\boldsymbol{\sigma}'$. Compute $a = \ell(\boldsymbol{\sigma}') - \ell(\boldsymbol{\sigma})$. Accept when $a \le 0$; when $a > 0$, accept with probability $e^{-\beta a}$, otherwise keep the current tour. Repeat and retain the best tour seen.

The proposal is symmetric. This Metropolis procedure has stationary distribution (1.5) for fixed β, and the acceptance rule uses a ratio of weights, so it does not require calculating Z_β. Accepting some worse routes allows the search to leave a local minimum. *Simulated annealing* increases β during the run: begin with more freedom to explore, then favour shorter routes more strongly. The cooling schedule and convergence rate determine what we can guarantee about the output.



$$\mathbf{M} = \begin{bmatrix}
0 & 2451 & 713 & 1018 & 1631 & 1374 & 2408 & 213 & 2571 & 875 & 1420 & 2145 & 1972 \\
2451 & 0 & 1745 & 1524 & 831 & 1240 & 959 & 2596 & 403 & 1589 & 1374 & 357 & 579 \\
713 & 1745 & 0 & 355 & 920 & 803 & 1737 & 851 & 1858 & 262 & 940 & 1453 & 1260 \\
1018 & 1524 & 355 & 0 & 700 & 862 & 1395 & 1123 & 1584 & 466 & 1056 & 1280 & 987 \\
1631 & 831 & 920 & 700 & 0 & 663 & 1021 & 1769 & 949 & 796 & 879 & 586 & 371 \\
1374 & 1240 & 803 & 862 & 663 & 0 & 1681 & 1551 & 1765 & 547 & 225 & 887 & 999 \\
2408 & 959 & 1737 & 1395 & 1021 & 1681 & 0 & 2493 & 678 & 1724 & 1891 & 1114 & 701 \\
213 & 2596 & 851 & 1123 & 1769 & 1551 & 2493 & 0 & 2699 & 1038 & 1605 & 2300 & 2099 \\
2571 & 403 & 1858 & 1584 & 949 & 1765 & 678 & 2699 & 0 & 1744 & 1645 & 653 & 600 \\
875 & 1589 & 262 & 466 & 796 & 547 & 1724 & 1038 & 1744 & 0 & 679 & 1272 & 1162 \\
1420 & 1374 & 940 & 1056 & 879 & 225 & 1891 & 1605 & 1645 & 679 & 0 & 1017 & 1200 \\
2145 & 357 & 1453 & 1280 & 586 & 887 & 1114 & 2300 & 653 & 1272 & 1017 & 0 & 504 \\
1972 & 579 & 1260 & 987 & 371 & 999 & 701 & 2099 & 600 & 1162 & 1200 & 504 & 0
\end{bmatrix}$$

Figure 1.5: A thirteen-city instance. The map shows the cities; the matrix $\mathbf{M}$ below gives the distances used to calculate tour lengths.

1.7.3 One Metropolis run

Figure 1.6 shows 1000 steps for the thirteen-city instance. The blue curve is the length $\ell(X_k)$ of the current tour. The red point marks the shortest tour found during this run, with length 7293.

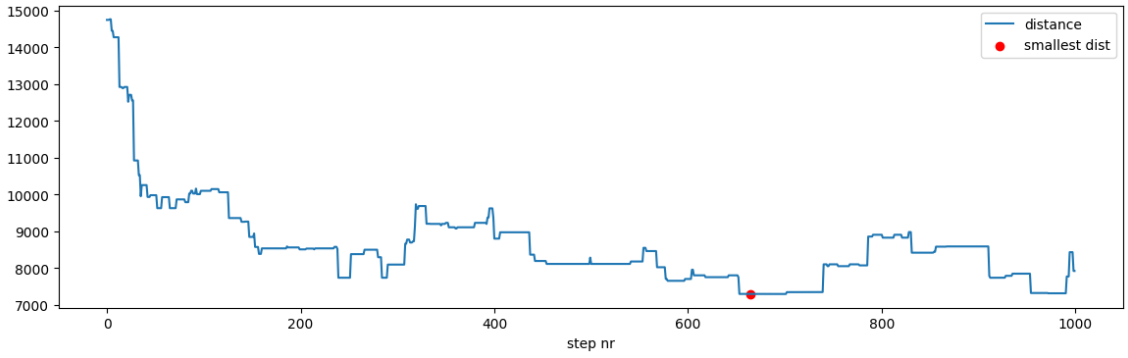


Figure 1.6: Tour lengths during one Metropolis run. The red point marks the best solution found in this run, not a certified global optimum.

The length sometimes increases: Metropolis can accept a worse tour, allowing the chain to explore other routes rather than only making improvements. We keep track of the best tour seen, even if the current tour later becomes longer. A single run illustrates the procedure; it does not guarantee that we have found the globally shortest tour.

1.8 Can we obtain an exact sample?

Perfect simulation is the art of turning a Markov chain with stationary law $\boldsymbol{\pi}$ into a sampling procedure that returns a state Y with

$$\mathbb{P}(Y = \mathbf{e}) = \pi(\mathbf{e}), \qquad \mathbf{e} \in \mathbb{E},$$

exactly. The procedure usually takes a random amount of time. Unlike a fixed-length MCMC run, its output has no remaining error from stopping before convergence. Top-to-random already illustrated one way to obtain such an exact sample.

There are several approaches. *Coupling from the past* (CFTP), due to Propp and Wilson [PW96], uses shared updates from the past. Another family of methods uses *strong stationary duality* (SSD) [DF90]: a suitable dual chain and its absorption time can provide a strong stationary time for the original chain. We will study the constructions and the assumptions under which they give practical exact samplers later. For now, the question is how much computation exactness costs compared with an approximate sample.

1.9 PageRank: ranking pages by following links

Suppose a collection of webpages contains links between them. How should we rank the pages? Counting incoming links is one idea, but a link from a widely visited page might matter more than a link from an obscure one. PageRank makes this recursive idea into a Markov-chain problem [PBMW98]. The basic PageRank score describes link-based importance across the collection; it does not depend on a particular search query. To rank results for a query, a search engine combines such a score with measures of relevance to that query [MRS08].

1.9.1 The random surfer

Let $\mathbb{E} = \{\mathbf{e}_1, \ldots, \mathbf{e}_M\}$ be the pages, and let X_k be the page visited after k moves. Fix $\alpha \in (0,1)$. At each move:

1. with probability α, follow an outgoing link chosen uniformly;
2. with probability $1-\alpha$, jump to a page chosen uniformly from all M pages, including the current one.

If a page has no outgoing links, choose the next page uniformly instead. This rule defines a Markov chain. We rank pages by their stationary probabilities: a larger $\pi(\mathbf{e}_i)$ means a larger long-run share of visits.

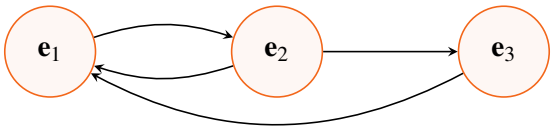
A page receives probability from the pages linking to it. A page with many outgoing links divides its contribution among them. For the following formula, treat a page with no outgoing links as having links to every page. If d_i is the resulting number of outgoing links from $\mathbf{e}_i$, stationarity says

$$\pi(\mathbf{e}_j) = \frac{1-\alpha}{M} + \alpha \sum_{i:\, \mathbf{e}_i \to \mathbf{e}_j} \frac{\pi(\mathbf{e}_i)}{d_i}. \tag{1.6}$$

The first term is the contribution of random jumps. The second is the contribution of incoming links. The score of a page depends on the scores of its neighbours, so we must determine all the scores together.

1.9.2 A three-page example

Imagine that the entire Internet consists of just three pages, with these links:



Page $\mathbf{e}_1$ has two incoming links, while $\mathbf{e}_2$ and $\mathbf{e}_3$ have one each. Take $\alpha = 1/2$, so half the moves follow links and half are uniform jumps. Equation (1.6) gives

$$\begin{aligned}
\pi(\mathbf{e}_1) &= \frac{1}{6} + \frac{1}{4}\pi(\mathbf{e}_2) + \frac{1}{2}\pi(\mathbf{e}_3),\\
\pi(\mathbf{e}_2) &= \frac{1}{6} + \frac{1}{2}\pi(\mathbf{e}_1),\\
\pi(\mathbf{e}_3) &= \frac{1}{6} + \frac{1}{4}\pi(\mathbf{e}_2).
\end{aligned}$$

Solving these equations yields

$$\boldsymbol{\pi} = \left(\frac{15}{39}, \frac{14}{39}, \frac{10}{39}\right).$$

Page $\mathbf{e}_1$ ranks first. Pages $\mathbf{e}_2$ and $\mathbf{e}_3$ have the same number of incoming links, but different scores: $\mathbf{e}_2$ receives all the link-following contribution of $\mathbf{e}_1$, whereas $\mathbf{e}_3$ shares that of $\mathbf{e}_2$ with another page. The directed arrows describe links; random jumps are possible between every pair of pages even though we have not drawn them.

What do we do with $\boldsymbol{\pi}$? The random surfer spends a long-run fraction $15/39$ of visits on $\mathbf{e}_1$, $14/39$ on $\mathbf{e}_2$, and $10/39$ on $\mathbf{e}_3$. PageRank uses these scores as a measure of the pages' importance, giving the link-based ranking $\mathbf{e}_1$, then $\mathbf{e}_2$, then $\mathbf{e}_3$.

A search engine must also take account of the user's question. For example, for the query "Markov chains", we need pages relevant to that topic. A high PageRank alone does not mean that a page answers the question: our highest-ranked page $\mathbf{e}_1$ might be about something else entirely. This is the idea behind PageRank in Google search: the link-based importance scores, which are independent of the query, can be combined with relevance to what the user is looking for. PageRank supplies an importance signal, rather than a complete rule for ordering search results.

1.10 A hidden Markov chain: weather seen through Bob's activities

1.10.1 We observe the activities, not the weather

Use the two-state weather model from the beginning of this chapter. Now suppose we do not observe the weather. Instead, each day we learn whether Bob jogged (J), went shopping (G), or watched television (T). His choices depend on that day's weather:

Weather	Jogging J	Shopping G	Television T
Sunny	0.80	0.10	0.10
Rainy	0.15	0.35	0.50

Given the whole weather sequence, Bob's daily choices are independent. For example, observing jogging makes sunshine more plausible, while television makes rain more plausible. But today's weather also gives information about tomorrow's weather, so the order of the observations matters.

Suppose Bob reports

```
JTGGTGJGTJJGTG.
```

To specify the model completely, let the probability that day zero is sunny be $3/4$, the stationary probability we calculated earlier. We can ask several different questions:

1. *How probable is this sequence of activities?* We need to add the probabilities over all possible weather sequences. The forward algorithm does this without listing every one of them.
2. *Which entire weather sequence is most probable given these activities?* The Viterbi algorithm finds a maximizing sequence. It keeps the best partial explanation ending in each weather state, then extends those explanations one day at a time.
3. *What is the probability of rain on one particular day, given all the activities?* The forward–backward algorithm combines observations before and after that day to calculate this probability.

The most probable full sequence can differ from the sequence obtained by choosing the most probable weather separately for each day. If the probabilities in the two tables are unknown, we can also try to learn them from the observations; this leads to algorithms such as Baum–Welch. We call the model a *hidden Markov model*: the weather is a Markov chain, and the activities give indirect observations of its states.

1.10.2 From weather to GPS and other noisy observations

Imagine a car travelling through a road network. Its true road segment is hidden, and the GPS device gives noisy positions. Picking the nearest road for each measurement can make the car appear to jump between nearby roads. A hidden Markov model can use both the measurement error and the possible moves along the network to select a plausible route. Viterbi solves the corresponding most-probable-path problem; Newson and Krumm give a GPS map-matching example [NK09].

We will meet the same type of question in other settings: which words could have produced a recorded sound, or which transmitted message could have produced a noisy signal? Viterbi also plays a major role in decoding convolutional codes in communications. Spacecraft approaching the

ISS must likewise infer their position and velocity from noisy measurements. Those continuous-state navigation problems often use Kalman or extended Kalman filters [NAS20]; the finite-state weather model provides an accessible starting point for the broader idea of inferring a hidden evolving state.

1.11 The questions we will develop

The examples lead to three tasks. First, understand the chain: what are its states and moves, can it reach all relevant states, and does it approach stationary probabilities independent of the start? Next, estimate how many steps it takes. Finally, use those estimates to justify a sampling, search, or inference algorithm.

Chapter 2 develops the finite-state theory and proves convergence in two ways. Chapter 3 studies its speed: we will run two chains together to bound a meeting time, use linear algebra to estimate decay rates, and construct strong stationary times such as the card-shuffling rule above. The later topics advertised here include MCMC, stochastic optimization, perfect simulation, ruin and duality, and hidden Markov models. Their full chapters remain to be developed. The final `tmp` chapter is a collection of examples for checking the notation and layout.

Chapter 2

Finite Markov chains

2.1 The Markov property and a transition matrix

Let $\mathbb{E} = \{\mathbf{e}_1, \dots, \mathbf{e}_M\}$ be a finite state space. Its elements may be numbers, vectors, permutations, or configurations. We use indices i, j for matrix calculations and state names $\mathbf{e}, \mathbf{e}'$ when the particular labels do not matter.

Definition 2.1.1 — Time-homogeneous Markov chain. A process $\mathbf{X} = (X_n)_{n \geq 0}$ with values in $\mathbb{E}$ is a time-homogeneous Markov chain with transition matrix $\mathbf{P} = (\mathbf{P}(\mathbf{e}_i, \mathbf{e}_j))_{i,j=1}^{M}$ if

$$\mathbb{P}(X_{n+1} = \mathbf{e}_j \mid X_0 = \mathbf{e}_{i_0}, \dots, X_n = \mathbf{e}_i) = \mathbf{P}(\mathbf{e}_i, \mathbf{e}_j) \tag{2.1}$$

for every history whose conditioning event has positive probability. The notation $\mathbf{P}(\mathbf{e}_i, \mathbf{e}_j)$ denotes the (i, j)th entry of $\mathbf{P}$. The entries satisfy $\mathbf{P}(\mathbf{e}, \mathbf{e}') \geq 0$ and $\sum_{\mathbf{e}'} \mathbf{P}(\mathbf{e}, \mathbf{e}') = 1$ for every $\mathbf{e}$.

The condition says that the present state contains all the information from the past needed to predict the next step. Time homogeneity means that the same matrix applies at every time. It does not mean that the distribution of X_n is the same at every time.

For individual entries we also use the scalar notation

$$P(i, j) = p_{ij} = \mathbf{P}(\mathbf{e}_i, \mathbf{e}_j).$$

The matrix itself is bold: $\mathbf{P} = (p_{ij})_{i,j=1}^{M}$.

The initial distribution is a row vector

$$\boldsymbol{\mu} = (\mu(\mathbf{e}_1), \dots, \mu(\mathbf{e}_M)), \qquad \mu(\mathbf{e}) = \mathbb{P}(X_0 = \mathbf{e}).$$

We write $\mathbf{X} \sim (\boldsymbol{\mu}, \mathbf{P})$. These two objects determine every finite path probability:

$$\mathbb{P}(X_0 = \mathbf{e}_{i_0}, \dots, X_n = \mathbf{e}_{i_n}) = \mu(\mathbf{e}_{i_0}) \prod_{k=0}^{n-1} \mathbf{P}(\mathbf{e}_{i_k}, \mathbf{e}_{i_{k+1}}). \tag{2.2}$$

This follows by successive conditioning and (2.1).

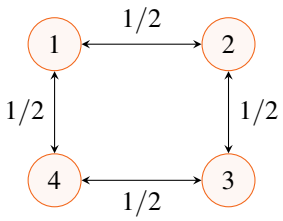
Definition 2.1.2 — A chain started from a specified state. $\mathbb{P}_{\mathbf{e}_i}$ denotes probabilities when the chain starts at $X_0 = \mathbf{e}_i$, with the same transition matrix $\mathbf{P}$. If $\mu(\mathbf{e}_i) > 0$, we can write

$$\mathbb{P}_{\mathbf{e}_i}(X_n = \mathbf{e}_j) = \mathbb{P}(X_n = \mathbf{e}_j \mid X_0 = \mathbf{e}_i).$$

Thus the subscript specifies the starting state. We use $\mathbb{E}_{\mathbf{e}_i}$ for expectation under this law.

Even if the original law has $\mu(\mathbf{e}_i) = 0$, $\mathbb{P}_{\mathbf{e}_i}$ is defined: we replace that initial law by a certain start at $\mathbf{e}_i$, keeping the same transition matrix $\mathbf{P}$.

Example 2.1 — A walk on four vertices. Arrange the states $1,2,3,4$ in a cycle. At each step choose one of the two neighbours uniformly. In that state order,



$$\mathbf{P} = \begin{pmatrix} 0 & 1/2 & 0 & 1/2 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 1/2 & 0 & 1/2 & 0 \end{pmatrix}.$$

If $X_0 = 1$, then $\boldsymbol{\mu} = (1,0,0,0)$, and

$$\boldsymbol{\mu}\mathbf{P} = (0,1/2,0,1/2), \qquad \boldsymbol{\mu}\mathbf{P}^2 = (1/2,0,1/2,0).$$

The state alternates between $\{1,3\}$ and $\{2,4\}$. This will become our first example of periodicity.

Remark 2.1.1 — Rows and columns. Distributions multiply matrices from the left. A scalar function $f : \mathbb{E} \to \mathbb{R}$ is represented by the column vector $\mathbf{f} = (f(\mathbf{e}_1), \dots, f(\mathbf{e}_M))^{\mathsf{T}}$. Then

$$\begin{aligned} (\mathbf{P}\mathbf{f})_i &= \sum_j \mathbf{P}(\mathbf{e}_i, \mathbf{e}_j) f(\mathbf{e}_j) \\ &= \sum_j f(\mathbf{e}_j) \mathbb{P}(X_1 = \mathbf{e}_j \mid X_0 = \mathbf{e}_i) = \mathbb{E}_{\mathbf{e}_i} f(X_1). \end{aligned}$$

For the constant function $f \equiv 1$, this gives $\mathbf{P}\mathbf{1} = \mathbf{1}$.

2.2 Distributions after several steps

Write $\boldsymbol{\mu}^{(n)}$ for the distribution of X_n, with $\boldsymbol{\mu}^{(0)} = \boldsymbol{\mu}$.

Proposition 2.2.1 — Evolution of the distribution. For every $n \geq 0$,

$$\boldsymbol{\mu}^{(n)} = \boldsymbol{\mu}\mathbf{P}^n. \tag{2.3}$$

Moreover, $\mathbf{P}^n(\mathbf{e}, \mathbf{e}') = \mathbb{P}_{\mathbf{e}}(X_n = \mathbf{e}')$, where $\mathbb{P}_{\mathbf{e}}$ means that $X_0 = \mathbf{e}$.

Proof. Condition on X_n and use the Markov property:

$$\begin{aligned} \mu^{(n+1)}(\mathbf{e}_j) &= \mathbb{P}(X_{n+1} = \mathbf{e}_j) \\ &= \sum_i \mathbb{P}(X_{n+1} = \mathbf{e}_j, X_n = \mathbf{e}_i) \\ &= \sum_i \mathbb{P}(X_{n+1} = \mathbf{e}_j \mid X_n = \mathbf{e}_i) \mathbb{P}(X_n = \mathbf{e}_i) \\ &= \sum_i \mathbb{P}(X_n = \mathbf{e}_i) \mathbf{P}(\mathbf{e}_i, \mathbf{e}_j) \\ &= \sum_i \mu^{(n)}(\mathbf{e}_i) \mathbf{P}(\mathbf{e}_i, \mathbf{e}_j). \end{aligned}$$

Terms whose conditioning event has probability zero contribute zero. Thus $\boldsymbol{\mu}^{(n+1)} = \boldsymbol{\mu}^{(n)}\mathbf{P}$, and induction gives (2.3). Starting with a point mass at $\mathbf{e}$ gives the interpretation of the row indexed by $\mathbf{e}$ of $\mathbf{P}^n$. ∎

Corollary 2.2.2 — Chapman–Kolmogorov equations. For $m, n \geq 0$,

$$\mathbf{P}^{m+n}(\mathbf{e}, \mathbf{e}') = \sum_{\mathbf{e}'' \in \mathbb{E}} \mathbf{P}^m(\mathbf{e}, \mathbf{e}'')\mathbf{P}^n(\mathbf{e}'', \mathbf{e}').$$

These equations can be read as matrix multiplication or as conditioning on the intermediate state at time m.

Example 2.2 — A two-state weather model. Use the state order rainy, sunny and the transition matrix:

$$\mathbf{P} = \begin{pmatrix} 1/2 & 1/2 \\ 1/10 & 9/10 \end{pmatrix}. \tag{2.4}$$

If $r_n = \mathbb{P}(X_n = \text{rainy})$, then

$$r_{n+1} = \tfrac{1}{2} r_n + \tfrac{1}{10}(1 - r_n) = \tfrac{1}{10} + \tfrac{2}{5} r_n.$$

The fixed point is $1/6$, so

$$r_n = \frac{1}{6} + \left(r_0 - \frac{1}{6}\right)\left(\frac{2}{5}\right)^n. \tag{2.5}$$

Starting from each possible state gives

$$\mathbf{P}^n = \begin{pmatrix} \frac{1}{6} + \frac{5}{6}(\frac{2}{5})^n & \frac{5}{6} - \frac{5}{6}(\frac{2}{5})^n \\ \frac{1}{6} - \frac{1}{6}(\frac{2}{5})^n & \frac{5}{6} + \frac{1}{6}(\frac{2}{5})^n \end{pmatrix}. \tag{2.6}$$

For an arbitrary initial law, let $\mathbb{P}(X_0 = \mathbf{e}_R) = r_0 = p$, so $\boldsymbol{\mu} = (p, 1-p)$. Multiplying by (2.6) gives

$$\begin{aligned} \mathbb{P}(X_n = \mathbf{e}_R) &= p\mathbf{P}^n(\mathbf{e}_R, \mathbf{e}_R) + (1-p)\mathbf{P}^n(\mathbf{e}_S, \mathbf{e}_R) \\ &= p\left[\frac{1}{6} + \frac{5}{6}\left(\frac{2}{5}\right)^n\right] + (1-p)\left[\frac{1}{6} - \frac{1}{6}\left(\frac{2}{5}\right)^n\right] \\ &= \frac{1}{6} + \left(p - \frac{1}{6}\right)\left(\frac{2}{5}\right)^n. \end{aligned}$$

Here $\mathbf{e}_R$ and $\mathbf{e}_S$ denote rainy and sunny. Thus

$$\boldsymbol{\mu}\mathbf{P}^n = \left(\frac{1}{6} + \left(p - \frac{1}{6}\right)\left(\frac{2}{5}\right)^n, \frac{5}{6} - \left(p - \frac{1}{6}\right)\left(\frac{2}{5}\right)^n\right).$$

Every initial distribution therefore approaches $(1/6, 5/6)$.

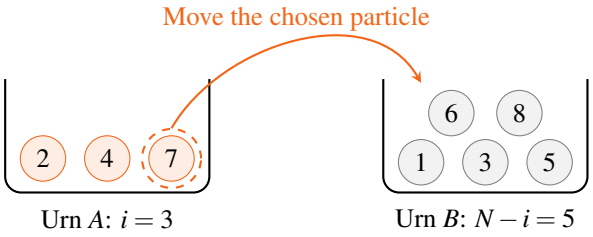
The same calculation has a linear algebra interpretation. For

$$\mathbf{A} = \begin{pmatrix} 1 & -5 \\ 1 & 1 \end{pmatrix}, \qquad \mathbf{D} = \begin{pmatrix} 1 & 0 \\ 0 & 2/5 \end{pmatrix}, \qquad \mathbf{A}^{-1} = \begin{pmatrix} 1/6 & 5/6 \\ -1/6 & 1/6 \end{pmatrix},$$

we have $\mathbf{P} = \mathbf{A}\mathbf{D}\mathbf{A}^{-1}$, and hence $\mathbf{P}^n = \mathbf{A}\mathbf{D}^n\mathbf{A}^{-1}$. The eigenvalue 1 determines the limiting part, while $(2/5)^n$ controls the disappearing part. Later chapters will turn this observation into general bounds on convergence.

2.3 Examples with large state spaces

Example 2.3 — The Ehrenfest urn model. There are N labelled particles, each in one of two urns, A and B. Choose a particle uniformly and move it to the other urn.



The picture has $N = 8$ and $i = 3$. Moving one of the three particles in A decreases the count; moving one of the five in B increases it. Let X_n be the number in urn A. Then $\mathbb{E} = \{0, 1, \ldots, N\}$ and

$$\mathbf{P}(i, i-1) = \frac{i}{N}, \qquad \mathbf{P}(i, i+1) = \frac{N-i}{N},$$

with only the applicable move at each boundary. The number in urn A is itself a Markov chain: these probabilities depend only on that number, not on which particles are in the urn.

There is also a more detailed description. Record which urn contains each particle, giving a configuration $\mathbf{e} \in \{0,1\}^N$. Flipping a uniformly chosen coordinate is a walk on the hypercube. Counting the ones recovers the Ehrenfest chain. This example shows that the choice of state affects both the size of the state space and the information retained by the model.

Example 2.4 — Top-to-random on three cards. Represent a deck by a permutation, from top to bottom. From $(1,2,3)$, reinserting the top card in positions $1, 2, 3$ produces $(1,2,3)$, $(2,1,3)$, $(2,3,1)$, each with probability $1/3$. In the state order

$$(123, 132, 213, 231, 312, 321),$$

the full matrix is

$$\mathbf{P} = \frac{1}{3}\begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

Every row and every column sums to one. For a large deck, we describe the operation directly instead of constructing the $n!$ by $n!$ matrix.

Exercise 2.1 Check the six rows of the matrix in Example 2.4 by listing the three possible outcomes of one move. Starting from 123, compute the probability of being at 321 after two steps. ∎

2.4 Simulating a chain with uniform random variables

A matrix specifies probabilities. A simulation needs a deterministic rule which takes a state and a random input and returns the next state.

Definition 2.4.1 — Initialisation and update functions. An initialisation function $\psi_\mu : [0,1] \to \mathbb{E}$ realises $\boldsymbol{\mu}$ if $\psi_\mu(U) \sim \boldsymbol{\mu}$ for $U \sim \mathscr{U}(0,1)$. An update function $\phi : \mathbb{E} \times [0,1] \to \mathbb{E}$ realises $\mathbf{P}$ if, for every $\mathbf{e}, \mathbf{e}' \in \mathbb{E}$,

$$\mathbb{P}(\phi(\mathbf{e}, U) = \mathbf{e}') = \mathbf{P}(\mathbf{e}, \mathbf{e}'). \tag{2.7}$$

Such functions always exist on a finite state space. Define cumulative row sums $c_0(\mathbf{e}) = 0$ and $c_j(\mathbf{e}) = \sum_{i=1}^{j} \mathbf{P}(\mathbf{e}, \mathbf{e}_i)$, and set

$$\phi(\mathbf{e}, u) = \mathbf{e}_j \quad \text{if } c_{j-1}(\mathbf{e}) \le u < c_j(\mathbf{e}).$$

Use the same construction with the entries of $\boldsymbol{\mu}$ to obtain ψ_μ. The value at $u = 1$ can be assigned arbitrarily, since it has probability zero. Zero-length intervals cause no difficulty. Explicitly, for a fixed row i,

$$\phi(\mathbf{e}_i, u) = \begin{cases} \mathbf{e}_1, & 0 \le u < \mathbf{P}(\mathbf{e}_i, \mathbf{e}_1), \\ \mathbf{e}_2, & \mathbf{P}(\mathbf{e}_i, \mathbf{e}_1) \le u < \mathbf{P}(\mathbf{e}_i, \mathbf{e}_1) + \mathbf{P}(\mathbf{e}_i, \mathbf{e}_2), \\ \vdots & \vdots \\ \mathbf{e}_M, & \displaystyle\sum_{j=1}^{M-1} \mathbf{P}(\mathbf{e}_i, \mathbf{e}_j) \le u < 1. \end{cases}$$

The interval assigned to $\mathbf{e}_j$ has length $\mathbf{P}(\mathbf{e}_i, \mathbf{e}_j)$, which proves (2.7). Half-open intervals specify each endpoint unambiguously; the choice at $u = 1$ does not change any probability.

Algorithm 2.1 — Simulation of a finite Markov chain. Generate independent $U_0, U_1, \ldots \sim \mathscr{U}(0,1)$ and set

$$X_0 = \psi_\mu(U_0), \qquad X_{n+1} = \phi(X_n, U_{n+1}).$$

Then $\mathbf{X} \sim (\boldsymbol{\mu}, \mathbf{P})$.

Proof. The initial law is correct by construction. Given the past up to time n, U_{n+1} is still uniform and independent. Equation (2.7) therefore supplies exactly the transition probability in (2.1). ■

Example 2.5 — A shared update for the weather chain. For (2.4), use

$$\phi(\text{rainy}, u) = \begin{cases} \text{rainy}, & u < 1/2, \\ \text{sunny}, & u \ge 1/2, \end{cases}$$

$$\phi(\text{sunny}, u) = \begin{cases} \text{rainy}, & u < 1/10, \\ \text{sunny}, & u \ge 1/10. \end{cases}$$

If two chains currently have different states but use the same u, they agree at the next step when $u < 1/10$ or $u \ge 1/2$. This occurs with probability $3/5$. Once they agree, the shared updates keep them together. This is our first concrete coupling construction.

Remark 2.4.1 The update function is not unique. Different rules can realise exactly the same matrix while behaving differently when several chains share a random input. That freedom is important for coupling and perfect simulation.

2.5 Accessibility and communicating classes

Definition 2.5.1 — Accessibility and communication. A state $\mathbf{e}'$ is accessible from $\mathbf{e}$, written $\mathbf{e} \to \mathbf{e}'$, if $\mathbf{P}^n(\mathbf{e},\mathbf{e}') > 0$ for some $n \geq 0$. States communicate, written $\mathbf{e} \leftrightarrow \mathbf{e}'$, if each is accessible from the other. A chain is irreducible if, for every $\mathbf{e}_i, \mathbf{e}_j \in \mathbb{E}$, there is an integer $n_{ij} \geq 1$ with

$$\mathbb{P}_{\mathbf{e}_i}(X_{n_{ij}} = \mathbf{e}_j) = \mathbf{P}^{n_{ij}}(\mathbf{e}_i,\mathbf{e}_j) > 0.$$

In a finite chain this is equivalent to all states communicating.

We allow $n = 0$ in accessibility so that each state is accessible from itself. Communication is an equivalence relation: reflexivity is immediate, symmetry is built into the definition, and transitivity follows from Chapman–Kolmogorov. Its equivalence classes are the communicating classes.

Definition 2.5.2 — Closed class and absorbing state. A communicating class $C \subseteq \mathbb{E}$ is closed if $\mathbf{P}(\mathbf{e},\mathbf{e}') = 0$ whenever $\mathbf{e} \in C$ and $\mathbf{e}' \notin C$. A state $\mathbf{a}$ is absorbing if $\mathbf{P}(\mathbf{a},\mathbf{a}) = 1$.

In the directed graph of positive transition probabilities, communication means mutual reachability and a closed class has no outgoing edge.

Example 2.6 — Two terminal classes. Consider

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \mathbb{E} = \{0,1,2\}.$$

The classes are $\{0\},\{1\},\{2\}$. The first and last are closed and absorbing. Starting at 1, the chain enters one of them after one step. No path leads back from either boundary to 1.

Proposition 2.5.1 Every finite chain has at least one closed communicating class. From every state, some closed class is accessible.

Proof. Collapse each communicating class to a vertex. The resulting graph has no directed cycle involving distinct classes, since such a cycle would make them communicate. A finite acyclic directed graph has a terminal vertex reachable from any given vertex. Terminal vertices correspond to closed classes. ∎

2.6 Periods and the obstruction to convergence

Definition 2.6.1 — Period. For a state with at least one possible return, define

$$d(\mathbf{e}) = \gcd\{n \geq 1 : \mathbf{P}^n(\mathbf{e},\mathbf{e}) > 0\}.$$

The state is aperiodic if $d(\mathbf{e}) = 1$. A chain is aperiodic if every state has period one: $d(\mathbf{e}_i) = 1$ for all $\mathbf{e}_i \in \mathbb{E}$.

In a finite irreducible chain every state has a possible return. If $\mathbf{P}(\mathbf{e},\mathbf{e}) > 0$, then $d(\mathbf{e}) = 1$, since one is an allowed return time. A self-loop is sufficient for aperiodicity, but not necessary.

Lemma 2.6.1 — Period is constant within a class. If $\mathbf{e}$ and $\mathbf{e}'$ communicate and have possible returns, then $d(\mathbf{e}) = d(\mathbf{e}')$.

Proof. Choose paths $\mathbf{e} \to \mathbf{e}'$ and $\mathbf{e}' \to \mathbf{e}$ of lengths r and s. Their concatenation is a return to $\mathbf{e}$ of length $r+s$. Let $n \geq 1$ be any integer such that $\mathbf{P}^n(\mathbf{e}',\mathbf{e}') > 0$. Then a return to $\mathbf{e}$ in $r+n+s$ steps is also possible. The period $d(\mathbf{e})$ divides both $r+s$ and $r+n+s$, hence their difference n. It divides

every such n, so it divides $d(\mathbf{e}')$. Interchanging $\mathbf{e}$ and $\mathbf{e}'$ proves equality. ■

Example 2.7 — Period two. The four-cycle of Example 2.1 has period two: returns are possible at time two, and all return times are even. For every $N \geq 1$, the non-lazy Ehrenfest model also has period two. Each move changes the count by one and hence changes its parity, so all returns have even length. A two-step return is possible: choose a particle and then choose that same particle again, an event of probability $1/N$. Repeating such two-step returns gives every positive even return length. Thus $d(i) = 2$ for all $i = 0, \ldots, N$, including the boundary counts. Irreducibility alone does not remove this obstruction.

Example 2.8 — A self-loop is not necessary for aperiodicity. A simple random walk on a triangle has no self-loop. Nevertheless, a state can be revisited after two steps or three steps, so its period is $\gcd(2,3) = 1$. Aperiodicity concerns all possible return lengths, not only the possibility of remaining in place for one step.

Exercise 2.2 Show that top-to-random shuffling on n cards is irreducible and aperiodic. For accessibility, consider the order in which the original top cards can be placed into prescribed final positions. For aperiodicity, identify a move which leaves the deck unchanged. ■

2.7 Eventually every transition is possible

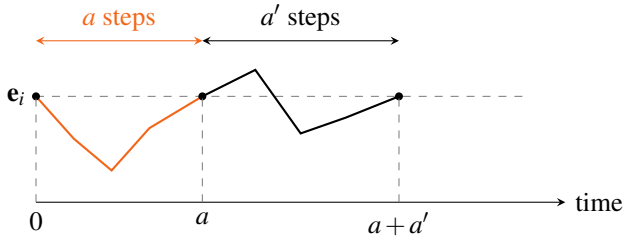
The next result turns structural assumptions into a uniform finite-time fact. We will use it in both proofs of the convergence theorem.

Lemma 2.7.1 — Return lengths with greatest common divisor one. Let A be a nonempty set of positive integers closed under addition. If $\gcd A = 1$, then A contains all sufficiently large integers.

We use this arithmetic lemma without proof; see Brémaud [Br99]. Its content is that, once the available lengths have gcd one, adding them together produces every sufficiently large length.

For a state $\mathbf{e}_i$, put $\mathscr{A}_i = \{n \geq 1 : \mathbf{P}^n(\mathbf{e}_i, \mathbf{e}_i) > 0\}$. If $a, a' \in \mathscr{A}_i$, then the Markov property gives

$$\begin{aligned}
\mathbf{P}^{a+a'}(\mathbf{e}_i, \mathbf{e}_i) &= \mathbb{P}_{\mathbf{e}_i}(X_{a+a'} = \mathbf{e}_i) \\
&\geq \mathbb{P}_{\mathbf{e}_i}(X_a = \mathbf{e}_i, X_{a+a'} = \mathbf{e}_i) \\
&= \mathbb{P}_{\mathbf{e}_i}(X_a = \mathbf{e}_i)\mathbb{P}(X_{a+a'} = \mathbf{e}_i \mid X_a = \mathbf{e}_i) \\
&= \mathbf{P}^a(\mathbf{e}_i, \mathbf{e}_i)\mathbf{P}^{a'}(\mathbf{e}_i, \mathbf{e}_i) > 0.
\end{aligned}$$



Here a, a' are numbers of steps. Concatenating these two returns proves that $\mathscr{A}_i$ is closed under addition. If the state is aperiodic, its return lengths have gcd one, so Lemma 2.7.1 gives an L_i such that $\mathbf{P}^n(\mathbf{e}_i, \mathbf{e}_i) > 0$ for all $n \geq L_i$. Taking $L = \max_i L_i$ works for every state at once.

Theorem 2.7.2 — Positive-power theorem. For a finite irreducible aperiodic chain, there is $m \geq 1$ such that

$$\mathbf{P}^n(\mathbf{e}, \mathbf{e}') > 0 \qquad \text{for all } \mathbf{e}, \mathbf{e}' \in \mathbb{E} \text{ and all } n \geq m. \tag{2.8}$$

Proof. The preceding argument gives L such that $\mathbf{P}^l(\mathbf{e}_j, \mathbf{e}_j) > 0$ for every j and every $l \geq L$. By irreducibility, for each pair i, j choose a length $r_{ij} \geq 1$ with $\mathbf{P}^{r_{ij}}(\mathbf{e}_i, \mathbf{e}_j) > 0$. For any $n \geq r_{ij} + L$, go from $\mathbf{e}_i$ to $\mathbf{e}_j$ in r_{ij} steps and then make a return to $\mathbf{e}_j$ of length $n - r_{ij}$. Consequently,

$$\mathbf{P}^n(\mathbf{e}_i, \mathbf{e}_j) \geq \mathbf{P}^{r_{ij}}(\mathbf{e}_i, \mathbf{e}_j)\mathbf{P}^{n-r_{ij}}(\mathbf{e}_j, \mathbf{e}_j) > 0.$$

There are finitely many pairs. Therefore $m = L + \max_{i,j} r_{ij}$ works for all pairs and all $n \geq m$. ■

Definition 2.7.1 A finite chain is called **ergodic** if it is irreducible and aperiodic.

The stationary-distribution existence theorem below needs only irreducibility; aperiodicity is needed for convergence at every deterministic time.

2.8 Hitting times have finite expectation

Let $T_{\mathbf{e}'} = \inf\{n \geq 0 : X_n = \mathbf{e}'\}$ and $T^+_{\mathbf{e}'} = \inf\{n \geq 1 : X_n = \mathbf{e}'\}$. The second quantity is a return time when the chain starts at $\mathbf{e}'$. We write $\mathbb{E}_{\mathbf{e}}$ for expectation conditional on $X_0 = \mathbf{e}$.

Lemma 2.8.1 — A geometric bound in time blocks. In a finite irreducible chain there are $m \geq 1$ and $\varepsilon > 0$, depending on $\mathbf{e}'$, such that

$$\mathbb{P}_{\mathbf{e}}(T_{\mathbf{e}'} > km) \leq (1 - \varepsilon)^k, \qquad \mathbf{e} \in \mathbb{E},\ k \geq 0.$$

In particular, $T_{\mathbf{e}'}$ is almost surely finite and $\mathbb{E}_{\mathbf{e}} T_{\mathbf{e}'} < \infty$. The return time $T^+_{\mathbf{e}'}$ also has finite expectation under $\mathbb{P}_{\mathbf{e}'}$.

Proof. We first find a uniform positive chance of hitting the target within one block. For every starting state $\mathbf{e}_i$, irreducibility supplies a time $n_i \geq 1$ with $\mathbf{P}^{n_i}(\mathbf{e}_i, \mathbf{e}') > 0$. Different states may require different times. Define

$$m = \max_i n_i, \qquad \varepsilon = \min_i \mathbf{P}^{n_i}(\mathbf{e}_i, \mathbf{e}') > 0.$$

Finiteness of the state space makes m finite and the minimum positive. For each i,

$$\{X_{n_i} = \mathbf{e}'\} \subseteq \{T_{\mathbf{e}'} \leq n_i\} \subseteq \{T_{\mathbf{e}'} \leq m\}.$$

Consequently,

$$\mathbb{P}_{\mathbf{e}_i}(T_{\mathbf{e}'} \leq m) \geq \mathbb{P}_{\mathbf{e}_i}(X_{n_i} = \mathbf{e}') = \mathbf{P}^{n_i}(\mathbf{e}_i, \mathbf{e}') \geq \varepsilon,$$
$$\mathbb{P}_{\mathbf{e}_i}(T_{\mathbf{e}'} > m) \leq 1 - \varepsilon.$$

This works even for a periodic chain: the hit can occur anywhere within the block, and need not occur at the same time for every starting state.

Iterating the block bound. The chance of avoiding the target for one block is at most $1 - \varepsilon$, from every starting state. Condition on survival through the first block and on the state at its end. By the

Markov property,

$$\begin{aligned}\mathbb{P}_{\mathbf{e}}(T_{\mathbf{e}'} > 2m) &= \sum_i \mathbb{P}_{\mathbf{e}}(T_{\mathbf{e}'} > m, X_m = \mathbf{e}_i)\mathbb{P}_{\mathbf{e}_i}(T_{\mathbf{e}'} > m)\\ &\le (1-\varepsilon)\sum_i \mathbb{P}_{\mathbf{e}}(T_{\mathbf{e}'} > m, X_m = \mathbf{e}_i)\\ &= (1-\varepsilon)\mathbb{P}_{\mathbf{e}}(T_{\mathbf{e}'} > m) \le (1-\varepsilon)^2.\end{aligned}$$

The state at the block boundary can depend on having avoided the target; the estimate still holds because it is uniform over all states. More generally,

$$\begin{aligned}\mathbb{P}_{\mathbf{e}}(T_{\mathbf{e}'} > (k+1)m) &= \sum_i \mathbb{P}_{\mathbf{e}}(T_{\mathbf{e}'} > km, X_{km} = \mathbf{e}_i)\mathbb{P}_{\mathbf{e}_i}(T_{\mathbf{e}'} > m)\\ &\le (1-\varepsilon)\sum_i \mathbb{P}_{\mathbf{e}}(T_{\mathbf{e}'} > km, X_{km} = \mathbf{e}_i)\\ &= (1-\varepsilon)\mathbb{P}_{\mathbf{e}}(T_{\mathbf{e}'} > km).\end{aligned}$$

Starting from $\mathbb{P}_{\mathbf{e}}(T_{\mathbf{e}'} > 0) \le 1$, induction proves the geometric bound. Using the tail-sum formula and grouping times into blocks,

$$\begin{aligned}\mathbb{E}_{\mathbf{e}} T_{\mathbf{e}'} &= \sum_{n\ge 0} \mathbb{P}_{\mathbf{e}}(T_{\mathbf{e}'} > n)\\ &= \sum_{k\ge 0} \sum_{n=km}^{(k+1)m-1} \mathbb{P}_{\mathbf{e}}(T_{\mathbf{e}'} > n)\\ &\le \sum_{k\ge 0} \sum_{n=km}^{(k+1)m-1} (1-\varepsilon)^k\\ &= m\sum_{k\ge 0}(1-\varepsilon)^k = \frac{m}{\varepsilon} < \infty.\end{aligned}$$

For a return from $\mathbf{e}'$, first take one step. Then

$$\mathbb{E}_{\mathbf{e}'} T^+_{\mathbf{e}'} = 1 + \sum_i \mathbf{P}(\mathbf{e}', \mathbf{e}_i)\mathbb{E}_{\mathbf{e}_i} T_{\mathbf{e}'} \le 1 + \frac{m}{\varepsilon} < \infty.$$

■

2.9 Stationary does not mean motionless

Definition 2.9.1 — Stationary distribution. A probability row vector $\boldsymbol{\pi}$ is stationary for $\mathbf{P}$ if

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}. \tag{2.9}$$

Equivalently, $\pi(\mathbf{e}') = \sum_{\mathbf{e}} \pi(\mathbf{e})\mathbf{P}(\mathbf{e}, \mathbf{e}')$ for every $\mathbf{e}' \in \mathbb{E}$.

If $X_0 \sim \boldsymbol{\pi}$, Proposition 2.2.1 gives $X_n \sim \boldsymbol{\pi}$ for every n. The chain still moves; it is the distribution which remains unchanged. To find a stationary distribution by linear algebra, solve (2.9) together with $\sum_{\mathbf{e}} \pi(\mathbf{e}) = 1$ and $\pi(\mathbf{e}) \ge 0$. The equations in (2.9) alone do not impose normalisation.

Example 2.9 — Stationarity on a cycle. For the four-cycle, $\boldsymbol{\pi} = (1/4, 1/4, 1/4, 1/4)$ is stationary. Nevertheless, starting at vertex 1 gives alternating distributions on the two parity classes. This illustrates the distinction between existence of equilibrium and convergence towards equilibrium from a specified starting state.

Example 2.10 — Many stationary distributions. For Example 2.6, every vector $\boldsymbol{\pi} = (a, 0, 1-a)$ with $0 \le a \le 1$ is stationary. Its mass is concentrated on the two absorbing states. The balance equations cannot select a unique weight between distinct closed classes.

2.10 Return cycles construct a stationary distribution

We construct equilibrium from occupation counts before a return. This gives a probabilistic meaning to the solution of the balance equations.

Start at $\mathbf{e}_1$ and let $\tau = T_{\mathbf{e}_1}^+$. For $i = 1, \dots, M$, define

$$s_i = \mathbb{E}_{\mathbf{e}_1}\left[\sum_{n=0}^{\tau-1} \mathbf{1}_{\{X_n=\mathbf{e}_i\}}\right] = \sum_{n\ge 0} \mathbb{P}_{\mathbf{e}_1}(X_n = \mathbf{e}_i, \tau > n). \tag{2.10}$$

These are the expected numbers of visits before the first return to $\mathbf{e}_1$.

Theorem 2.10.1 — Existence and the return-cycle formula. Every finite irreducible chain has a strictly positive stationary distribution:

$$\boldsymbol{\pi} = \frac{1}{\mathbb{E}_{\mathbf{e}_1}\tau}(s_1, \dots, s_M), \qquad \pi(\mathbf{e}_i) = \frac{s_i}{\mathbb{E}_{\mathbf{e}_1}\tau}. \tag{2.11}$$

In particular, $\pi(\mathbf{e}_1) = 1/\mathbb{E}_{\mathbf{e}_1} T_{\mathbf{e}_1}^+$.

Proof. First, Lemma 2.8.1 gives $\mathbb{E}_{\mathbf{e}_1}\tau < \infty$, and $0 \le s_i \le \mathbb{E}_{\mathbf{e}_1}\tau$. At every time before τ, the chain occupies exactly one state, so

$$\begin{aligned}\sum_i s_i &= \mathbb{E}_{\mathbf{e}_1}\left[\sum_{n=0}^{\tau-1}\sum_i \mathbf{1}_{\{X_n=\mathbf{e}_i\}}\right] \\ &= \mathbb{E}_{\mathbf{e}_1}\left[\sum_{n=0}^{\tau-1} 1\right] = \mathbb{E}_{\mathbf{e}_1}\tau.\end{aligned}$$

Also $s_1 = 1$: time zero is the only visit to $\mathbf{e}_1$ before its first return. For $i \ne 1$, choose a shortest path from $\mathbf{e}_1$ to $\mathbf{e}_i$. It cannot return to $\mathbf{e}_1$ on the way, and its probability is positive. Hence $s_i > 0$. Thus the proposed vector has positive entries and sums to one.

To prove stationarity, fix j. Expand each occupation count by (2.10) and use the Markov property:

$$\begin{aligned}\sum_i s_i \mathbf{P}(\mathbf{e}_i, \mathbf{e}_j) &= \sum_i \sum_{n\ge 0} \mathbb{P}_{\mathbf{e}_1}(X_n = \mathbf{e}_i, \tau > n)\mathbf{P}(\mathbf{e}_i, \mathbf{e}_j) \\ &= \sum_{n\ge 0}\sum_i \mathbb{P}_{\mathbf{e}_1}(X_n = \mathbf{e}_i, X_{n+1} = \mathbf{e}_j, \tau > n) \\ &= \sum_{n\ge 0} \mathbb{P}_{\mathbf{e}_1}(X_{n+1} = \mathbf{e}_j, \tau > n) \\ &= \mathbb{E}_{\mathbf{e}_1}\sum_{n=1}^{\tau} \mathbf{1}_{\{X_n=\mathbf{e}_j\}}.\end{aligned}$$

The event $\{\tau > n\}$ is determined by the history through time n; we keep it unchanged when taking the next step. All summands are nonnegative, so interchanging sums is valid. Finally, $X_0 = X_\tau = \mathbf{e}_1$ implies, on every trajectory,

$$\sum_{n=1}^{\tau} \mathbf{1}_{\{X_n=\mathbf{e}_j\}} - \sum_{n=0}^{\tau-1} \mathbf{1}_{\{X_n=\mathbf{e}_j\}} = \mathbf{1}_{\{X_\tau=\mathbf{e}_j\}} - \mathbf{1}_{\{X_0=\mathbf{e}_j\}}.$$

Since both endpoints are $\mathbf{e}_1$,

$$\mathbf{1}_{\{X_\tau=\mathbf{e}_j\}} - \mathbf{1}_{\{X_0=\mathbf{e}_j\}} = \begin{cases} 1-1=0, & \text{if } j=1, \\ 0-0=0, & \text{if } j\neq 1. \end{cases}$$

The shifted sum therefore has expectation s_j. We have proved $\mathbf{sP}=\mathbf{s}$ for $\mathbf{s}=(s_1,\dots,s_M)$; division by $\mathbb{E}_{\mathbf{e}_1}\tau$ gives $\boldsymbol{\pi}\mathbf{P}=\boldsymbol{\pi}$. ■

Theorem 2.10.2 — Uniqueness under irreducibility. A finite irreducible chain has exactly one stationary distribution. Aperiodicity is not required.

Proof. Let $\boldsymbol{\pi}$ be the positive stationary distribution constructed above and let $\boldsymbol{\nu}$ be any other one. Put $c=\min_{\mathbf{e}'}\nu(\mathbf{e}')/\pi(\mathbf{e}')$ and $\mathbf{h}=\boldsymbol{\nu}-c\boldsymbol{\pi}$. Then $\mathbf{h}$ is nonnegative, stationary, and has at least one zero entry, say $h(\mathbf{e}'')=0$. If $h(\mathbf{e})>0$ for some $\mathbf{e}$, irreducibility gives an n with $\mathbf{P}^n(\mathbf{e},\mathbf{e}'')>0$. But

$$0=h(\mathbf{e}'')=(\mathbf{h}\mathbf{P}^n)(\mathbf{e}'')\geq h(\mathbf{e})\mathbf{P}^n(\mathbf{e},\mathbf{e}'')>0,$$

This would say $0>0$, which is impossible. The contradiction disproves our assumption that $h(\mathbf{e})>0$ at some state. Since all entries of $\mathbf{h}$ are nonnegative, they must all be zero. Hence $\boldsymbol{\nu}=c\boldsymbol{\pi}$. Both probability vectors sum to one, so $1=c\cdot 1$ and $c=1$. Thus $\boldsymbol{\nu}=\boldsymbol{\pi}$, proving uniqueness. ■

Corollary 2.10.3 — Mean return time. For every state $\mathbf{e}'$ of a finite irreducible chain,

$$\mathbb{E}_{\mathbf{e}'}T^+_{\mathbf{e}'}=\frac{1}{\pi(\mathbf{e}')}.$$

Proof. For $\mathbf{e}'=\mathbf{e}_1$, this is exactly the construction in Theorem 2.10.1: since $s_1=1$, $\pi(\mathbf{e}_1)=1/\mathbb{E}_{\mathbf{e}_1}T^+_{\mathbf{e}_1}$. For any other $\mathbf{e}'$, repeat the same construction with $\mathbf{e}'$ in place of $\mathbf{e}_1$: start at $\mathbf{e}'$ and stop at its first return time $T^+_{\mathbf{e}'}$. Before that return, the cycle contains exactly one visit to $\mathbf{e}'$, at time zero. The resulting stationary distribution therefore assigns probability $1/\mathbb{E}_{\mathbf{e}'}T^+_{\mathbf{e}'}$ to $\mathbf{e}'$. By uniqueness, this distribution is the same $\boldsymbol{\pi}$, proving the formula. ■

2.11 Finding equilibrium in examples

Definition 2.11.1 — Detailed balance. A probability distribution $\boldsymbol{\pi}$ satisfies detailed balance for $\mathbf{P}$ if

$$\pi(\mathbf{e})\mathbf{P}(\mathbf{e},\mathbf{e}')=\pi(\mathbf{e}')\mathbf{P}(\mathbf{e}',\mathbf{e}),\qquad \mathbf{e},\mathbf{e}'\in\mathbb{E}.$$

The chain is then reversible with respect to $\boldsymbol{\pi}$.

Summing over $\mathbf{e}$ proves stationarity, since $\sum_{\mathbf{e}}\pi(\mathbf{e})\mathbf{P}(\mathbf{e},\mathbf{e}')=\pi(\mathbf{e}')\sum_{\mathbf{e}}\mathbf{P}(\mathbf{e}',\mathbf{e})=\pi(\mathbf{e}')$. Detailed balance is a convenient sufficient condition, not a necessary one.

Example 2.11 — Uniformity for a doubly stochastic matrix. If columns as well as rows sum to one, the uniform distribution $\pi(\mathbf{e})=1/M$ is stationary. This applies to the cycle, the hypercube walk, and top-to-random shuffling. For the last example, each target deck has n possible predecessors, obtained by moving one of its cards to the top; each contributes probability $1/n$ to the column sum.

Example 2.12 — The binomial law in the Ehrenfest model. For Example 2.3, it turns out that the stationary distribution is $\mathrm{Bin}(N, 1/2)$:

$$\pi(i) = 2^{-N}\binom{N}{i}, \qquad 0 \le i \le N.$$

Checking stationarity is an exercise on the problem list. By irreducibility, this stationary distribution is unique. The model is still periodic. Replacing $\mathbf{P}$ by the lazy matrix $(\mathbf{I}+\mathbf{P})/2$ preserves $\boldsymbol{\pi}$ and removes that periodicity.

Exercise 2.3 For the weather chain (2.4), solve the balance equations to recover $\boldsymbol{\pi} = (1/6, 5/6)$. Use Corollary 2.10.3 to find the mean return time to a rainy day and the mean return time to a sunny day. Remember that the return time $T_{\mathbf{e}'}^{+}$ counts at least one step. ∎

2.12 Convergence to stationarity

We now know how to construct the stationary distribution and when it is unique. The next question concerns a chain started away from equilibrium.

Theorem 2.12.1 — Finite-state convergence theorem. Let $\mathbf{P}$ be irreducible and aperiodic on the finite state space $\mathbb{E}$. For every initial distribution $\boldsymbol{\mu}$,

$$\boldsymbol{\mu}\mathbf{P}^n \longrightarrow \boldsymbol{\pi} \qquad (n \to \infty), \tag{2.12}$$

where $\boldsymbol{\pi}$ is the unique stationary distribution. Equivalently, $\mathbf{P}^n(\mathbf{e}, \mathbf{e}') \to \pi(\mathbf{e}')$ for all $\mathbf{e}, \mathbf{e}' \in \mathbb{E}$.

We couple a chain started from $\boldsymbol{\mu}$ with a stationary chain, then bound the total variation distance by the probability that they have not yet met. We use the positive-power Theorem 2.7.2.

2.12.1 Constructing the coupling

Start two independent chains $\mathbf{X}$ and $\mathbf{X}'$ with transition matrix $\mathbf{P}$, with $X_0 \sim \boldsymbol{\mu}$ and $X_0' \sim \boldsymbol{\pi}$. Their joint process $\mathbf{Z} = (Z_n)_{n \ge 0}$, where $Z_n = (X_n, X_n')$, has state space $\mathbb{E} \times \mathbb{E}$ and transition matrix $\mathbf{Q}$ given by

$$\mathbf{Q}((\mathbf{e}_i, \mathbf{e}_j), (\mathbf{e}_k, \mathbf{e}_l)) = \mathbf{P}(\mathbf{e}_i, \mathbf{e}_k)\mathbf{P}(\mathbf{e}_j, \mathbf{e}_l).$$

Independence also gives

$$\mathbf{Q}^n((\mathbf{e}_i, \mathbf{e}_j), (\mathbf{e}_k, \mathbf{e}_l)) = \mathbf{P}^n(\mathbf{e}_i, \mathbf{e}_k)\mathbf{P}^n(\mathbf{e}_j, \mathbf{e}_l). \tag{2.13}$$

For a sufficiently large n, both factors are positive. Thus the product chain is irreducible. This is where we use aperiodicity of the original chain: two irreducible periodic chains need not be able to meet at the same time.

Let

$$T = \inf\{n \ge 0 : X_n = X_n'\}$$

be the hitting time of the diagonal $\Delta = \{(\mathbf{e}, \mathbf{e}) : \mathbf{e} \in \mathbb{E}\}$. Fix $\mathbf{a} \in \mathbb{E}$. Since $\mathbf{Q}$ is a finite irreducible transition matrix, Lemma 2.8.1 implies that its hitting time of $(\mathbf{a}, \mathbf{a})$ has finite expectation from every starting pair. The chain hits Δ no later, so $T < \infty$ almost surely.

We can also bound the meeting time directly. Take m from Theorem 2.7.2 and set $\lambda = \min_{i,j} \mathbf{P}^m(\mathbf{e}_i, \mathbf{e}_j) > 0$. For any starting pair,

$$\begin{aligned}\mathbb{P}(X_m = \mathbf{a}, X'_m = \mathbf{a} \mid X_0 = \mathbf{e}_i, X'_0 = \mathbf{e}_j) &= \mathbf{P}^m(\mathbf{e}_i, \mathbf{a})\mathbf{P}^m(\mathbf{e}_j, \mathbf{a}) \\ &\geq \lambda^2.\end{aligned}$$

For independent arbitrary initial laws this is $\mathbb{P}(X_m = \mathbf{a})\mathbb{P}(X'_m = \mathbf{a}) \geq \lambda^2$. Being together at time m implies $T \leq m$. At later block boundaries, condition on the current pair and the event that no meeting has occurred yet. The fresh future random inputs are independent, so the same bound applies to every current pair. Consequently,

$$\begin{aligned}\mathbb{P}(T > (k+1)m) &\leq (1-\lambda^2)\mathbb{P}(T > km), \\ \mathbb{P}(T > km) &\leq (1-\lambda^2)^k.\end{aligned}$$

The current pair may be dependent after conditioning on $T > km$; this does not affect independence of the future updates conditional on that pair.

Define a new process $\mathbf{X}''$ by following $\mathbf{X}$ until T and then following $\mathbf{X}'$:

$$X''_n = \begin{cases} X_n, & n < T, \\ X'_n, & n \geq T. \end{cases}$$

At time T the two states agree. Given the joint history through any time, the next step of $\mathbf{X}''$ has row $\mathbf{P}(X''_n, \cdot)$: on $\{T > n\}$, use the next step of $\mathbf{X}$ (if the chains meet at $n+1$, their states there coincide); on $\{T \leq n\}$, use the next step of $\mathbf{X}'$. The independent fresh random inputs give row $\mathbf{P}(X''_n, \cdot)$ in either case. Conditioning further only on the history of $\mathbf{X}''$ keeps that same row. Hence $\mathbf{X}''$ is a Markov chain with initial law $\boldsymbol{\mu}$ and transition matrix $\mathbf{P}$. Also, $X'_n \sim \boldsymbol{\pi}$ for every n.

Remark 2.12.1 — The role of the meeting time. At a deterministic time n, the coupled chains can still differ if $T > n$. The argument bounds the probability of that event. It does not assert that the chain has the stationary law conditional on the event $\{T \leq n\}$. Strong stationary times, introduced in Section 3.6, require a stronger independence property.

2.12.2 Total variation distance

We introduce the part of total variation theory needed for the convergence proof. Chapter 3 develops its use in quantitative bounds.

Definition 2.12.1 — Total variation distance. For probability distributions $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ on $\mathbb{E}$, put

$$d_{TV}(\boldsymbol{\mu}, \boldsymbol{\nu}) = \frac{1}{2} \sum_{\mathbf{e} \in \mathbb{E}} |\mu(\mathbf{e}) - \nu(\mathbf{e})|. \tag{2.14}$$

Lemma 2.12.2 — Event probabilities. For probability distributions $\boldsymbol{\mu}, \boldsymbol{\nu}$,

$$d_{TV}(\boldsymbol{\mu}, \boldsymbol{\nu}) = \max_{A \subseteq \mathbb{E}} |\mu(A) - \nu(A)|. \tag{2.15}$$

Proof. Exercise. ■

2.12.3 Convergence in total variation

Proof of Theorem 2.12.1. Use the chains $\mathbf{X}'$ and $\mathbf{X}''$ constructed above, with $X_n' \sim \boldsymbol{\pi}$ and $X_n'' \sim \boldsymbol{\mu}^{(n)}$. The event formula gives

$$\begin{aligned} d_{TV}(\boldsymbol{\mu}^{(n)}, \boldsymbol{\pi}) &= \max_{A \subseteq \mathbb{E}} |\mu^{(n)}(A) - \pi(A)| \\ &= \max_{A \subseteq \mathbb{E}} |\mathbb{P}(X_n'' \in A) - \mathbb{P}(X_n' \in A)|. \end{aligned}$$

For each A, split the two probabilities according to whether the other chain belongs to A:

$$\begin{aligned} \mathbb{P}(X_n'' \in A) &= \mathbb{P}(X_n'' \in A, X_n' \in A) + \mathbb{P}(X_n'' \in A, X_n' \notin A), \\ \mathbb{P}(X_n' \in A) &= \mathbb{P}(X_n'' \in A, X_n' \in A) + \mathbb{P}(X_n'' \notin A, X_n' \in A). \end{aligned}$$

The first terms cancel. Therefore

$$\begin{aligned} |\mathbb{P}(X_n'' \in A) - \mathbb{P}(X_n' \in A)| &= |\mathbb{P}(X_n'' \in A, X_n' \notin A) - \mathbb{P}(X_n'' \notin A, X_n' \in A)| \\ &\le \mathbb{P}(X_n'' \in A, X_n' \notin A) + \mathbb{P}(X_n'' \notin A, X_n' \in A) \\ &\le \mathbb{P}(X_n'' \neq X_n') \le \mathbb{P}(T > n). \end{aligned}$$

The two disagreement events are disjoint and both require different states. The last bound does not depend on A. Taking the maximum proves

$$d_{TV}(\boldsymbol{\mu}^{(n)}, \boldsymbol{\pi}) \le \mathbb{P}(T > n) \longrightarrow 0.$$

This proves convergence in total variation, and hence convergence at every state. ■

Corollary 2.12.3 — A uniform geometric bound. Choose m from Theorem 2.7.2 and put $\lambda = \min_{i,j} \mathbf{P}^m(\mathbf{e}_i, \mathbf{e}_j) > 0$. For every initial law and $n \ge 0$,

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}^n, \boldsymbol{\pi}) \le (1 - \lambda^2)^{\lfloor n/m \rfloor}. \tag{2.16}$$

Proof. For $k = \lfloor n/m \rfloor$, $\mathbb{P}(T > n) \le \mathbb{P}(T > km) \le (1-\lambda^2)^k$. Combine this with the bound from the convergence proof. If $\lambda = 1$, interpret the bound as one for $n < m$ and zero for $n \ge m$. ■

The bound in (2.16) can be far from sharp; we next seek estimates reflecting the particular dynamics.

2.13 What fails without the assumptions?

Example 2.13 — Periodic equilibrium. For $\mathbf{P} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, the stationary distribution is $(1/2, 1/2)$, but a chain started at 0 alternates deterministically. Its even-time and odd-time laws do not converge to the same limit. Starting in the stationary distribution does give the stationary law at every time.

Example 2.14 — Dependence on the starting state. In Example 2.6, a chain started at 0 stays there and a chain started at 2 stays there. There is no single limiting law which works for every initial distribution. The different closed classes retain information about the start forever.

Exercise 2.4 Suppose $\mathbf{P}$ is irreducible. Show that its lazy version $\mathbf{P}_{\mathrm{L}} = (\mathbf{I} + \mathbf{P})/2$ is irreducible and aperiodic, and has the same stationary distribution. Apply Theorem 2.12.1 to the lazy four-cycle and the lazy Ehrenfest model. ■

2.14 From convergence to a running-time estimate

Theorem 2.12.1 guarantees convergence for any finite irreducible aperiodic chain. A sampling algorithm also needs a running-time estimate: how large should n be to make the error smaller than ε? For the weather chain we can read the answer from (2.5). For a deck with $n!$ states, calculating all entries of $\mathbf{P}^k$ is impractical. Chapter 3 develops three approaches: coupling, spectral estimates and strong stationary times. Each gives a way to bound the error without computing the full transition matrix power.

Chapter 3

Rate of convergence to stationarity

Chapter 2 proves that finite irreducible aperiodic chains converge. We now estimate how long this takes. Throughout this chapter, $\boldsymbol{\pi}$ denotes the stationary distribution and the state space is $\mathbb{E}$. We begin with coupling, then use spectral decomposition and bounds on the spectral gap, and finally construct strong stationary times.

3.1 Total variation distance and coupling

3.1.1 Total variation as an error in event probabilities

Recall Definition 2.12.1 and Lemma 2.12.2:

$$d_{TV}(\boldsymbol{\mu},\boldsymbol{\nu}) = \frac{1}{2}\sum_{\mathbf{e}\in\mathbb{E}} |\mu(\mathbf{e}) - \nu(\mathbf{e})| = \max_{A\subseteq\mathbb{E}} |\mu(A) - \nu(A)|.$$

Thus $d_{TV}(\boldsymbol{\mu},\boldsymbol{\nu}) \le \varepsilon$ means that the probability of any event differs by at most ε under the two laws.

Exercise 3.1 Show that

$$d_{TV}(\boldsymbol{\mu},\boldsymbol{\nu}) = \sum_{\mathbf{e}:\mu(\mathbf{e})>\nu(\mathbf{e})} (\mu(\mathbf{e}) - \nu(\mathbf{e})) = 1 - \sum_{\mathbf{e}} \min\{\mu(\mathbf{e}), \nu(\mathbf{e})\}.$$

■

Lemma 3.1.1 — A stochastic matrix cannot increase total variation. For every stochastic matrix $\mathbf{K}$,

$$d_{TV}(\boldsymbol{\mu}\mathbf{K}, \boldsymbol{\nu}\mathbf{K}) \le d_{TV}(\boldsymbol{\mu},\boldsymbol{\nu}). \tag{3.1}$$

Proof. Expand the matrix products directly:

$$\begin{aligned}
d_{TV}(\boldsymbol{\mu}\mathbf{K}, \boldsymbol{\nu}\mathbf{K}) &= \frac{1}{2}\sum_{\mathbf{e}'} |(\boldsymbol{\mu}\mathbf{K})(\mathbf{e}') - (\boldsymbol{\nu}\mathbf{K})(\mathbf{e}')| \\
&= \frac{1}{2}\sum_{\mathbf{e}'} \left|\sum_{\mathbf{e}} (\mu(\mathbf{e}) - \nu(\mathbf{e}))\mathbf{K}(\mathbf{e},\mathbf{e}')\right| \\
&\le \frac{1}{2}\sum_{\mathbf{e}'}\sum_{\mathbf{e}} |\mu(\mathbf{e}) - \nu(\mathbf{e})|\mathbf{K}(\mathbf{e},\mathbf{e}') \\
&= \frac{1}{2}\sum_{\mathbf{e}} |\mu(\mathbf{e}) - \nu(\mathbf{e})| \sum_{\mathbf{e}'} \mathbf{K}(\mathbf{e},\mathbf{e}') \\
&= \frac{1}{2}\sum_{\mathbf{e}} |\mu(\mathbf{e}) - \nu(\mathbf{e})| = d_{TV}(\boldsymbol{\mu},\boldsymbol{\nu}).
\end{aligned}$$

We use the triangle inequality, nonnegative matrix entries, and the fact that every row of $\mathbf{K}$ sums to one. ■

In particular, stationarity gives

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}^{n+1},\boldsymbol{\pi}) = d_{TV}\big((\boldsymbol{\mu}\mathbf{P}^{n})\mathbf{P},\boldsymbol{\pi}\mathbf{P}\big) \le d_{TV}(\boldsymbol{\mu}\mathbf{P}^{n},\boldsymbol{\pi}).$$

Once the distance is below a chosen tolerance, subsequent steps keep it below that tolerance.

3.1.2 Couplings of probability distributions

Definition 3.1.1 — Coupling. A coupling of $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ is a pair of random variables (X,Y) on one probability space with marginal laws $X \sim \boldsymbol{\mu}$ and $Y \sim \boldsymbol{\nu}$. Its joint probability mass function $w(\mathbf{e},\mathbf{e}')$ satisfies

$$\sum_{\mathbf{e}'} w(\mathbf{e},\mathbf{e}') = \mu(\mathbf{e}), \qquad \sum_{\mathbf{e}} w(\mathbf{e},\mathbf{e}') = \nu(\mathbf{e}').$$

We write $\mathbf{w} = (w(\mathbf{e},\mathbf{e}'))_{\mathbf{e},\mathbf{e}'\in\mathbb{E}}$ for the joint probability matrix.

Theorem 3.1.2 — Coupling inequality and an optimal coupling. For any coupling (X,Y) of $\boldsymbol{\mu},\boldsymbol{\nu}$,

$$d_{TV}(\boldsymbol{\mu},\boldsymbol{\nu}) \le \mathbb{P}(X \neq Y).$$

There is a coupling attaining equality.

Proof. For any $A \subseteq \mathbb{E}$,

$$|\mathbb{P}(X \in A) - \mathbb{P}(Y \in A)| \le \mathbb{P}(X \neq Y).$$

Maximise over A. To attain equality, put $a(\mathbf{e}) = \min\{\mu(\mathbf{e}),\nu(\mathbf{e})\}$ and $s = \sum_{\mathbf{e}} a(\mathbf{e})$. If $s = 1$, the laws coincide; set $X = Y$ with that law. If $s < 1$, define

$$w(\mathbf{e},\mathbf{e}') = \mathbf{1}_{\{\mathbf{e}=\mathbf{e}'\}}a(\mathbf{e}) + \frac{(\mu(\mathbf{e}) - a(\mathbf{e}))(\nu(\mathbf{e}') - a(\mathbf{e}'))}{1-s}.$$

Both marginals are correct. On the diagonal the second term is zero, since at least one factor vanishes. Hence $\mathbb{P}(X = Y) = s$, and Exercise 3.1 gives $\mathbb{P}(X \neq Y) = 1 - s = d_{TV}(\boldsymbol{\mu},\boldsymbol{\nu})$. ■

3.1.3 Coupling chains and bounding a mixing time

A coupling of chains $\mathbf{X} \sim (\boldsymbol{\mu},\mathbf{P})$ and $\mathbf{Y} \sim (\boldsymbol{\nu},\mathbf{P})$ is a joint construction with the prescribed marginal path laws. A Markovian coupling chooses the next pair from the current pair and gives each coordinate its prescribed transition row. We normally arrange that once the chains meet, they use the same moves. Then

$$T = \inf\{n \ge 0 : X_n = Y_n\}$$

is the coupling time and $\{X_n \neq Y_n\} = \{T > n\}$. Taking $\boldsymbol{\nu} = \boldsymbol{\pi}$ in Theorem 3.1.2 yields

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}^{n},\boldsymbol{\pi}) \le \mathbb{P}(T > n). \tag{3.2}$$

The meeting-time proof in Section 2.12.1 uses this observation without needing the full coupling theorem.

Proposition 3.1.3 — Optimality at a fixed time. For each fixed n, the laws $\boldsymbol{\mu}\mathbf{P}^n$ and $\boldsymbol{\pi}$ admit a coupling (U,V) with $\mathbb{P}(U \neq V) = d_{TV}(\boldsymbol{\mu}\mathbf{P}^n, \boldsymbol{\pi})$.

This follows from Theorem 3.1.2. It does not guarantee that one Markovian coupling of the paths attains equality at all times.

Define the worst-case distance and mixing time by

$$d(n) = \max_{\mathbf{e}\in\mathbb{E}} d_{TV}(\boldsymbol{\delta}_{\mathbf{e}}\mathbf{P}^n, \boldsymbol{\pi}), \tag{3.3}$$

$$\tau_{mix}^{TV}(\varepsilon) = \min\{n \geq 0 : d(n) \leq \varepsilon\}, \qquad \tau_{mix}^{TV} = \tau_{mix}^{TV}(1/4). \tag{3.4}$$

Here $\boldsymbol{\delta}_{\mathbf{e}}$ is the probability row vector concentrated at $\mathbf{e}$. For a specified initial law, we write $\tau_{mix}^{TV}(\boldsymbol{\mu},\varepsilon) = \min\{n : d_{TV}(\boldsymbol{\mu}\mathbf{P}^n, \boldsymbol{\pi}) \leq \varepsilon\}$. Stochastic contraction, (3.1), makes each of these distances non-increasing in n.

Exercise 3.2 If for every starting state $\mathbf{e}$ we have a coupling with a stationary chain whose coupling time $T_{\mathbf{e}}$ satisfies $\mathbb{E}T_{\mathbf{e}} \leq B$, show that

$$\tau_{mix}^{TV}(\varepsilon) \leq \lceil B/\varepsilon \rceil .$$

Use Markov's inequality in (3.2). ∎

We can also couple chains started at two arbitrary states. Put

$$\bar{d}(n) = \max_{\mathbf{e},\mathbf{e}'\in\mathbb{E}} d_{TV}(\mathbf{P}^n(\mathbf{e},\cdot), \mathbf{P}^n(\mathbf{e}',\cdot)).$$

Lemma 3.1.4 For every initial law $\boldsymbol{\mu}$, $d_{TV}(\boldsymbol{\mu}\mathbf{P}^n, \boldsymbol{\pi}) \leq d(n) \leq \bar{d}(n)$.

Proof. The first inequality follows by writing $\boldsymbol{\mu}\mathbf{P}^n$ as a mixture of its rows. For the second, stationarity gives $\boldsymbol{\pi} = \sum_{\mathbf{e}'} \pi(\mathbf{e}')\mathbf{P}^n(\mathbf{e}',\cdot)$, so convexity of the sum of absolute values gives

$$d_{TV}(\mathbf{P}^n(\mathbf{e},\cdot), \boldsymbol{\pi}) \leq \sum_{\mathbf{e}'} \pi(\mathbf{e}') d_{TV}(\mathbf{P}^n(\mathbf{e},\cdot), \mathbf{P}^n(\mathbf{e}',\cdot)) \leq \bar{d}(n).$$

■

We will use the same random choices for two starting states and bound their meeting time uniformly over those states.

3.1.4 Random-to-Top card shuffling

Assume we have n cards. Consider the following card shuffling (one step described):

> Select a card uniformly at random and put it on the top of the deck.

More formally, we consider a chain $\mathbf{X}$ on a state space $\mathbb{E} = \mathscr{S}_n$ of all permutations of n elements. For $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_n)$ define

$$f_i(\boldsymbol{\sigma}) = (\sigma_i, \sigma_1, \sigma_2, \ldots, \sigma_{i-1}, \sigma_{i+1}, \ldots, \sigma_n),$$

i.e., $f_i(\boldsymbol{\sigma})$ is a permutation obtained from $\boldsymbol{\sigma}$ by placing i-th element (card) "on the top". Note that $f_1(\boldsymbol{\sigma}) = \boldsymbol{\sigma}$. The transition matrix is then the following:

$$\mathbf{P}(\boldsymbol{\sigma},\boldsymbol{\sigma}') = \begin{cases} \frac{1}{n} & \text{if} \quad \boldsymbol{\sigma}' = f_i(\boldsymbol{\sigma}) \text{ for some } i \in \{1,\ldots,n\} \\ 0 & \text{otherwise} \end{cases}$$

Of course, the stationary distribution is uniform, i.e., $\pi(\boldsymbol{\sigma}) = \frac{1}{n!}$ for any $\boldsymbol{\sigma} \in \mathscr{S}_n$.

We construct a coupling $(\mathbf{X}, \mathbf{Y})$ in the following way. Assume $X_0 = \boldsymbol{\sigma}$ and $Y_0 = \boldsymbol{\sigma}'$. Given (X_k, Y_k) do the following:

- Choose a card i uniformly at random.
- Find card i in both permutations X_k and Y_k
- Move that card to the top in both decks. The positions of the chosen card can differ, so the indices in the two uses of f need not coincide.

Roughly speaking, we search for the same card in both decks, and put it on the top in each one. Obviously, marginally X_k and Y_k are both chains corresponding to Random-To-Top card shuffling. However, they are strongly dependent. After 1 step, surely the same card is on the top of both decks. Moreover, if each card is chosen at least once, then $X_k = Y_k$ (and then $X_t = Y_t$ for $t \geq k$). Let $T = \min_k \{X_k = Y_k\}$, it is stochastically dominated by the time for a coupon collector to collect all n coupons (see Exercise 3.4). Thus $\mathbb{P}(T > n\log n + cn) \leq e^{-c}$ (for any $c > 0$) and hence

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}^{\lceil n\log n + cn\rceil}, \boldsymbol{\pi}) \leq e^{-c}.$$

It means that we have to make $n\log n$ steps plus a linear factor cn more steps to be close to stationarity.

Exercise 3.3 Show that for a Random-to-Top card shuffling for total variation mixing time we have

$$\tau_{mix}^{TV}(\varepsilon) \leq \lceil n\log n + n\log(\varepsilon^{-1})\rceil$$

■

Exercise 3.4 — Coupon collector problem. At one step we draw a single coupon out of n coupons, independently from other drawings and uniformly at random (i.e., each coupon has probability $1/n$ of appearing). Let T be the first moment when each coupon was drawn at least once. Show that

$$\mathbb{P}(T > k) \leq e^{-c} \quad \text{for} \quad k = \lceil n\log n + cn\rceil.$$

■

3.1.5 Lazy symmetric random walk on the d-dimensional cube

Consider the state space of d dimensional 0-1 valued vectors $\mathbb{E} = \{0,1\}^d$. The *lazy* symmetric random walk is defined by a transition matrix

$$\mathbf{P}(\mathbf{e}, \mathbf{e}') = \begin{cases} \frac{1}{2} & \text{if} \quad \mathbf{e} = \mathbf{e}', \\ \frac{1}{2d} & \text{if} \quad \mathbf{e} = \mathbf{e}' \pm \mathbf{s}_i, i \in \{1,\ldots,d\} \\ 0 & \text{otherwise,} \end{cases},$$

where $\mathbf{s}_i = (0,\ldots,0,1,0,\ldots,0)$ with 1 at position i. Roughly speaking, being in $\mathbf{e} = (e_1,\ldots,e_d)$ the random walk either does nothing with prob. $1/2$ or otherwise it changes randomly chosen i-th coordinate from e_i to $1 - e_i$. Of course the stationary distribution is uniform $\pi(\mathbf{e}) = 1/2^d$.

The key for constructing the coupling is the observation, that the above t.m. $\mathbf{P}$ corresponds to the following procedure:

> Choose a coordinate uniformly at random (i.e., with prob. $1/d$) and then with probability 1/2 set it to 0 or 1.

I.e., we choose a coordinate uniformly at random, and regardless its value, we set it to 0 or 1 (with equal prob.). It leads directly to the following coupling: Assume at step k we have X_k and Y_k, then

- Choose a coordinate $i \in \{1,\dots,d\}$ uniformly at random.
- Flip a coin and set $R = 0$ if heads and $R = 1$ otherwise.
- Set i-th coordinate of both X_k and Y_k to R.

Again, if each coordinate were chosen at least once, we surely have $X_k = Y_k$. The time T when it happens is bounded by the same coupon-collector time as in Random-to-Top (Section 3.1.4) example, we thus have (for any $c > 0$)

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}^{\lceil d\log d + cd\rceil}, \boldsymbol{\pi}) \le e^{-c}.$$

3.1.6 The hypercube with holding probability $1/(d+1)$

Again let $\mathbb{E} = \{0,1\}^d$, but now choose i uniformly from $\{0,1,\dots,d\}$. If $i = 0$, do nothing; otherwise flip coordinate i. The transition probabilities are

$$\mathbf{P}(\mathbf{e},\mathbf{e}') = \begin{cases} 1/(d+1), & \mathbf{e}' = \mathbf{e} \text{ or } \mathbf{e}' \text{ differs in one coordinate,} \\ 0, & \text{otherwise.} \end{cases}$$

The stationary law is uniform. This walk has a self-loop but, for $d > 1$, it is not lazy in the convention $\mathbf{P}(\mathbf{e},\mathbf{e}) \ge 1/2$.

Here is a coupling that preserves both transition laws. Let D be the set of disagreeing coordinates of the current vectors $\mathbf{e},\mathbf{e}'$, and $r = |D|$. All coordinates not mentioned in an update stay unchanged.

- If $r = 0$, choose the same move for both chains.
- If $r > 1$, choose a cyclic permutation f of D with no fixed points and extend it by the identity to $\{0,\dots,d\} \setminus D$. Choose i uniformly from $\{0,\dots,d\}$. Flip coordinate i in the first chain and $f(i)$ in the second, with index zero meaning no move.
- If $r = 1$, write $D = \{j\}$ and use the permutation of move indices that swaps 0 and j and fixes all other indices. Again apply move i to the first chain and the permuted move to the second.

Each chain uses a uniformly chosen move index, so the marginals are correct. For $r > 1$, an index in D removes two disagreements, while other indices preserve r. The waiting time for $r \to r-2$ is geometric with success probability $r/(d+1)$. For $r = 1$, either $i = 0$ or $i = j$ removes the last disagreement, giving probability $2/(d+1)$.

By the strong Markov property the successive waiting times are independent. Starting from any pair, their sum T satisfies

$$\mathbb{E}T \le \frac{d+1}{2}\log d + O(d), \qquad \operatorname{Var} T = O(d^2).$$

For the mean, sum $(d+1)/r$ along the decreasing even or odd sequence of values of r, with the final one-disagreement stage handled separately. For the variance, use $\operatorname{Var}(\mathrm{Geo}(p)) = (1-p)/p^2$ and the convergence of $\sum_{r\ge1} r^{-2}$. Chebyshev's inequality and the coupling bound yield

$$\tau_{mix}^{TV}(\varepsilon) \le \frac{1}{2} d\log d + O_\varepsilon(d)$$

for any fixed $\varepsilon \in (0,1)$ as $d \to \infty$.

Remark 3.1.1 The sharper asymptotic estimate has leading term $\frac{1}{4} d\log d$ for this walk, cf. [DS81]. This coupling gives the correct order, with a factor two in the leading upper bound.

3.1.7 Random transpositions and the choice of coupling

Let us consider the following example:

Example 3.1 — Random Transposition Shuffle. Roughly speaking:

> Select two cards independently and uniformly at random. If cards are different, swap them.

More formally: consider a chain $\mathbf{X}$ on $\mathbb{E} = \mathscr{S}_n$ of all permutations of n elements. For $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_n)$ define

$$f_{i,j}(\boldsymbol{\sigma}) = (\sigma_1, \ldots, \sigma_{i-1}, \sigma_j, \sigma_{i+1}, \ldots, \sigma_{j-1}, \sigma_i, \sigma_{j+1}, \ldots, \sigma_n),$$

i.e., $f_{i,j}(\boldsymbol{\sigma})$ is a permutation obtained by swapping cards on positions i and j (and we understand that f_{ii} yields no change). We display the formula for $i < j$ and set $f_{j,i} = f_{i,j}$.

$$\mathbf{P}(\boldsymbol{\sigma}, \boldsymbol{\sigma}') = \begin{cases} \frac{1}{n} & \text{if} \quad \boldsymbol{\sigma}' = \boldsymbol{\sigma} \\ \frac{2}{n^2} & \text{if} \quad \boldsymbol{\sigma}' = f_{i,j}(\boldsymbol{\sigma}) \text{ for } i \neq j, i, j \in \{1, \ldots, n\} \\ 0 & \text{otherwise} \end{cases}$$

Note that the following procedure is equivalent (i.e., yielding the same transition matrix $\mathbf{P}$):

> Pick card c and position r uniformly at random, swap card c with card at position r

Using the latter description, the following coupling $(\mathbf{X}, \mathbf{Y})$ comes to mind: Assume $X_0 = \boldsymbol{\sigma}$ and $\mathbf{Y}_0 = \boldsymbol{\sigma}'$. Given (X_k, Y_k) do the following

- Pick card c and position r uniformly at random.
- In both decks, i.e. X_k and Y_k swap card c with card at position r

Note that the position in both decks is chosen the same, whereas card c can be in different positions in both decks. Assume that at some step k both decks X_k and Y_k differ only on two positions:

	X_k:	Y_k:
	17	17
	24	24
	⋮	
	11	11
position $i \to$	c	c'
	24	24
	⋮	
	1	1
position $j \to$	c'	c
	37	37
	⋮	

The decks meet if we choose either card c or c' and either position i or j. These are four equally likely choices out of n^2. Any other choice leaves two disagreeing positions, possibly at new locations. Thus the final waiting time $T_{c,c'}$ is geometric with parameter $4/n^2$, and $\mathbb{E}T_{c,c'} = n^2/4$. From a pair differing in two positions, $\mathbb{P}(T > k) = (1 - 4/n^2)^k$. This coupling therefore needs order n^2 steps for its final stage and cannot prove the actual $O(n \log n)$ mixing rate. For fixed error tolerance, the random-transposition shuffle has mixing time $\frac{1}{2} n \log n + O(n)$; see [DS81].

Theorem 3.1.2 gives an optimal coupling of the laws at a fixed time. Constructing one Markovian path coupling with the same accuracy is a separate problem. The coupling above illustrates how a natural joint update can have a slow final stage even when the chain itself mixes faster. Non-Markovian constructions can use information beyond the current pair; for random transpositions see [BK06].

3.1.8 Time reversal and shuffling on groups

For an irreducible chain with stationary distribution $\boldsymbol{\pi}$, define

$$\widetilde{\mathbf{P}}(\mathbf{e},\mathbf{e}') = \frac{\pi(\mathbf{e}')\mathbf{P}(\mathbf{e}',\mathbf{e})}{\pi(\mathbf{e})}. \tag{3.5}$$

We write either $\widetilde{\mathbf{P}}$ or $\overleftarrow{\mathbf{P}}$ for this transition matrix. The two notations refer to the same reversal. Its rows sum to one by stationarity of $\boldsymbol{\pi}$, and $\boldsymbol{\pi}$ is also stationary for $\widetilde{\mathbf{P}}$.

Lemma 3.1.5 — Reversing a stationary path. For any sequence $\mathbf{e}_{i_0},\dots,\mathbf{e}_{i_k}$,

$$\pi(\mathbf{e}_{i_0})\prod_{r=0}^{k-1}\mathbf{P}(\mathbf{e}_{i_r},\mathbf{e}_{i_{r+1}}) = \pi(\mathbf{e}_{i_k})\prod_{r=0}^{k-1}\widetilde{\mathbf{P}}(\mathbf{e}_{i_{r+1}},\mathbf{e}_{i_r}).$$

Proof. Substitute (3.5); the stationary weights telescope. ■

Thus a stationary path observed backwards has the law of the reversed chain. For a reversible chain $\widetilde{\mathbf{P}} = \mathbf{P}$.

Random walks on finite groups. A finite group $\mathcal{G}$ has an associative operation, an identity id and an inverse g^{-1} for each element. Given an increment distribution $\boldsymbol{\nu}$, a left random walk moves from g to hg with probability $\nu(h)$:

$$\mathbf{P}(g,g') = \nu(g'g^{-1}).$$

For each fixed g', summing over g gives a column sum of one. Hence the uniform law $\pi(g) = 1/|\mathcal{G}|$ is stationary. The reversal has increment law $\nu^{\leftarrow}(h) = \nu(h^{-1})$. The hypercube walks are walks on $\mathbb{Z}_2^d$: their increments are the zero vector and the coordinate vectors $\mathbf{s}_i$. For the second walk the law is uniform on $\{\mathbf{0},\mathbf{s}_1,\dots,\mathbf{s}_d\}$.

Lemma 3.1.6 For a random walk on a finite group,

$$d_{TV}(\boldsymbol{\delta}_{id}\mathbf{P}^k,\boldsymbol{\pi}) = d_{TV}(\boldsymbol{\delta}_{id}\widetilde{\mathbf{P}}^k,\boldsymbol{\pi}).$$

The same distance holds from any fixed starting element.

Proof. Starting at id, write $X_k = H_k\cdots H_1$ for independent increments with law $\boldsymbol{\nu}$. Then $X_k^{-1} = H_1^{-1}\cdots H_k^{-1}$ has the same law as a reversed walk at time k, because we can reverse the order of an independent identically distributed tuple. Inversion is a bijection of $\mathcal{G}$ preserving the uniform law, so it preserves TV distance from that law. Starting at g right-multiplies the walk by g, another such bijection. ■

Example 3.2 — Top-to-random is the reversal of random-to-top. Represent a deck by the permutation $\boldsymbol{\sigma} = (\sigma_1,\dots,\sigma_n)$. Random-to-top applies one of the n position permutations that moves a position to the top. Its inverse moves the top position back to that position. Thus the reversed walk is top-to-random: choose i uniformly from $\{1,\dots,n\}$ and use

$$g_i(\boldsymbol{\sigma}) = (\sigma_2,\dots,\sigma_i,\sigma_1,\sigma_{i+1},\dots,\sigma_n),$$

with g_1 interpreted as the identity. Since the stationary law is uniform, (3.5) also verifies this by transposing the transition matrix.

Example 3.3 — Transferring the shuffling bound. Combining the random-to-top coupling with Lemma 3.1.6, we obtain for top-to-random, from any initial law,

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}^{\lceil n\log n+cn\rceil},\boldsymbol{\pi}) \le e^{-c}, \qquad c>0. \tag{3.6}$$

First transfer the bound for each deterministic deck, then use convexity for an initial mixture. Section 3.6 obtains a bound directly for top-to-random using a strong stationary time.

3.2 Spectral decomposition and bounds from eigenvalues

In this section assume $|\mathbb{E}| = M \ge 2$; a one-state chain is already stationary. A stochastic matrix has the constant column vector $\mathbf{1}$ as a right eigenvector with eigenvalue one, and $\boldsymbol{\pi}$ as a left eigenvector:

$$\mathbf{P}\mathbf{1} = \mathbf{1}, \qquad \boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}.$$

For an ergodic chain, Theorem 2.12.1 says that $\mathbf{P}^n \to \mathbf{1}\boldsymbol{\pi}$. Eigenvalues describe the decay of the remaining modes.

3.2.1 Diagonalizable transition matrices

Theorem 3.2.1 — Perron–Frobenius for a regular stochastic matrix. If a finite stochastic matrix has a strictly positive power, the eigenvalue one is simple and all its other eigenvalues have modulus less than one.

Here we can deduce the statement from the convergence theorem. If $\mathbf{P}\mathbf{f} = \lambda\mathbf{f}$ and $\lambda \ne 1$, multiplying by $\boldsymbol{\pi}$ gives $\boldsymbol{\pi}\mathbf{f} = 0$. Then $\mathbf{P}^n\mathbf{f} \to \mathbf{1}\boldsymbol{\pi}\mathbf{f} = \mathbf{0}$, so $|\lambda| < 1$. Convergence of $\mathbf{P}^n$ also excludes a Jordan block at one, and its eigenspace consists of constant vectors.

Suppose now that $\mathbf{P}$ is diagonalizable over $\mathbb{C}$. Collect right eigenvectors as columns of an invertible matrix $\mathbf{N}$ and write the rows of $\mathbf{N}^{-1}$ as $\boldsymbol{\ell}_1,\dots,\boldsymbol{\ell}_M$. Choose $\mathbf{f}_1 = \mathbf{1}$ and $\boldsymbol{\ell}_1 = \boldsymbol{\pi}$. Then

$$\mathbf{P}\mathbf{f}_j = \lambda_j\mathbf{f}_j, \tag{3.7}$$
$$\boldsymbol{\ell}_i\mathbf{f}_j = \mathbf{1}_{\{i=j\}}. \tag{3.8}$$

Let $\mathbf{D} = \operatorname{diag}(1,\lambda_2,\dots,\lambda_M)$ and define

$$\mathbf{B}_j = \mathbf{f}_j\boldsymbol{\ell}_j. \tag{3.9}$$

These rank-one matrices satisfy

$$\mathbf{B}_i\mathbf{B}_j = \begin{cases} \mathbf{0}, & i \ne j,\\ \mathbf{B}_j, & i = j. \end{cases} \tag{3.10}$$

Thus

$$\mathbf{P}^n = \mathbf{N}\mathbf{D}^n\mathbf{N}^{-1} = \mathbf{1}\boldsymbol{\pi} + \sum_{j=2}^{M}\lambda_j^n\mathbf{B}_j. \tag{3.11}$$

We use

$$\lambda_* = \max_{2\le j\le M}|\lambda_j|$$

for the largest modulus among non-stationary eigenvalues. In this subsection the indices $2,\dots,M$ need not follow any real-number ordering.

Theorem 3.2.2 — Geometric bound for a diagonalizable matrix. For a finite ergodic diagonalizable transition matrix there is $C < \infty$ such that

$$|\mathbf{P}^n(\mathbf{e},\mathbf{e}') - \pi(\mathbf{e}')| \le C\lambda_*^n, \qquad n \ge 1.$$

Proof. By (3.11), the left-hand side is at most $\lambda_*^n \sum_{j=2}^{M} |B_j(\mathbf{e},\mathbf{e}')|$. Take the maximum of the finite sum over $\mathbf{e},\mathbf{e}'$. ■

Remark 3.2.1 Summing the entrywise estimate gives $d_{TV}(\boldsymbol{\delta}_\mathbf{e}\mathbf{P}^n,\boldsymbol{\pi}) \le MC\lambda_*^n/2$. For a non-diagonalizable matrix, Jordan blocks can introduce factors such as $n\lambda_*^n$. A bound with a constant times λ_*^n need not hold. For every $\rho \in (\lambda_*,1)$, however, a constant C_ρ gives a bound $C_\rho\rho^n$. The elementary bound in Corollary 2.12.3 already proves a geometric rate without diagonalizability.

3.2.2 Reversible chains and the two spectral gaps

For a real scalar function f on $\mathbb{E}$, let $\mathbf{f} = (f(\mathbf{e}))_{\mathbf{e}\in\mathbb{E}}$ be its column of values. Define

$$\langle \mathbf{f},\mathbf{g}\rangle_{\boldsymbol{\pi}} = \sum_{\mathbf{e}} \pi(\mathbf{e}) f(\mathbf{e}) g(\mathbf{e}), \qquad \|\mathbf{f}\|_{\boldsymbol{\pi}}^2 = \langle \mathbf{f},\mathbf{f}\rangle_{\boldsymbol{\pi}}.$$

Detailed balance makes $\mathbf{P}$ self-adjoint in this inner product. Equivalently, for $\mathbf{D}_\pi = \operatorname{diag}(\pi(\mathbf{e}) : \mathbf{e} \in \mathbb{E})$, the matrix $\mathbf{D}_\pi^{1/2}\mathbf{P}\mathbf{D}_\pi^{-1/2}$ is symmetric. Consequently, a reversible matrix has a real orthonormal eigenbasis and we now order its eigenvalues by their values:

$$1 = \lambda_1 > \lambda_2 \ge \cdots \ge \lambda_M \ge -1.$$

For $M \ge 2$, define

$$\gamma = 1 - \lambda_2, \qquad \lambda_* = \max\{\lambda_2, |\lambda_M|\}, \qquad \gamma_* = 1 - \lambda_*. \tag{3.12}$$

We call γ the spectral gap and γ_* the absolute spectral gap. The latter controls discrete-time convergence. Cheeger's and Poincaré's inequalities bound γ. For a lazy chain, $\mathbf{P}(\mathbf{e},\mathbf{e}) \ge 1/2$ for all $\mathbf{e}$, the eigenvalues are nonnegative and $\gamma = \gamma_*$. Indeed, $\mathbf{P} = (\mathbf{I}+\mathbf{K})/2$ for a stochastic matrix $\mathbf{K}$.

Corollary 3.2.3 — Reversible spectral bound in total variation. For a finite irreducible aperiodic reversible chain,

$$d_{TV}(\boldsymbol{\delta}_\mathbf{e}\mathbf{P}^n,\boldsymbol{\pi}) \le \frac{1}{2}\sqrt{\frac{1}{\pi(\mathbf{e})} - 1}\,\lambda_*^n \le \frac{\lambda_*^n}{2\sqrt{\pi(\mathbf{e})}}. \tag{3.13}$$

Proof. Put $\boldsymbol{\mu}_n = \boldsymbol{\delta}_\mathbf{e}\mathbf{P}^n$ and let $\boldsymbol{\rho}_n = (\mu_n(\mathbf{e}')/\pi(\mathbf{e}'))_{\mathbf{e}'\in\mathbb{E}}$ be the column of density values. Detailed balance gives $\boldsymbol{\rho}_{n+1} = \mathbf{P}\boldsymbol{\rho}_n$. The vector $\boldsymbol{\rho}_n - \mathbf{1}$ has $\boldsymbol{\pi}$-mean zero, so the orthonormal spectral decomposition yields

$$\|\boldsymbol{\rho}_n - \mathbf{1}\|_{\boldsymbol{\pi}} \le \lambda_*^n \|\boldsymbol{\rho}_0 - \mathbf{1}\|_{\boldsymbol{\pi}}.$$

Cauchy–Schwarz gives

$$d_{TV}(\boldsymbol{\mu}_n,\boldsymbol{\pi}) = \frac{1}{2}\sum_{\mathbf{e}'} \pi(\mathbf{e}')|\rho_n(\mathbf{e}') - 1| \le \frac{1}{2}\|\boldsymbol{\rho}_n - \mathbf{1}\|_{\boldsymbol{\pi}}.$$

Finally, $\|\boldsymbol{\rho}_0 - \mathbf{1}\|_{\boldsymbol{\pi}}^2 = 1/\pi(\mathbf{e}) - 1$. ■

Example 3.4 — Why a negative eigenvalue matters. For $0 < a < 1$, the matrix

$$\mathbf{P} = \begin{pmatrix} a & 1-a \\ 1-a & a \end{pmatrix}$$

has stationary distribution $(1/2, 1/2)$ and nontrivial eigenvalue $2a-1$. Starting at 0, its total variation distance is $|2a-1|^n/2$. For small a, the spectral gap $2(1-a)$ is large, but the absolute gap $2a$ is small: the chain almost alternates between the states.

3.2.3 The Dirichlet form and a variational formula

Write $\Lambda(\mathbf{e}, \mathbf{e}') = \pi(\mathbf{e})\mathbf{P}(\mathbf{e}, \mathbf{e}')$; reversibility gives $\Lambda(\mathbf{e}, \mathbf{e}') = \Lambda(\mathbf{e}', \mathbf{e})$. For columns of function values define

$$\begin{aligned} \mathscr{E}(\mathbf{f}, \mathbf{g}) &= \frac{1}{2} \sum_{\mathbf{e}, \mathbf{e}'} \Lambda(\mathbf{e}, \mathbf{e}')(f(\mathbf{e}) - f(\mathbf{e}'))(g(\mathbf{e}) - g(\mathbf{e}')) \\ &= \langle \mathbf{f}, (\mathbf{I} - \mathbf{P})\mathbf{g} \rangle_{\boldsymbol{\pi}}. \end{aligned} \tag{3.14}$$

The second equality follows by expanding and using the row sums, stationarity and reversibility. Also,

$$\mathrm{Var}_{\boldsymbol{\pi}}(f) = \sum_{\mathbf{e}} \pi(\mathbf{e}) f(\mathbf{e})^2 - \left(\sum_{\mathbf{e}} \pi(\mathbf{e}) f(\mathbf{e}) \right)^2 = \frac{1}{2} \sum_{\mathbf{e}, \mathbf{e}'} \pi(\mathbf{e}) \pi(\mathbf{e}') (f(\mathbf{e}) - f(\mathbf{e}'))^2. \tag{3.15}$$

We use $\mathscr{E}$ for the Dirichlet form and $\mathbb{E}$ for the state space.

Proposition 3.2.4 — Variational characterization of the spectral gap.

$$\gamma = 1 - \lambda_2 = \inf_{f \text{ nonconstant}} \frac{\mathscr{E}(\mathbf{f}, \mathbf{f})}{\mathrm{Var}_{\boldsymbol{\pi}}(f)}. \tag{3.16}$$

Proof. Subtract the $\boldsymbol{\pi}$-mean from $\mathbf{f}$; this changes neither numerator nor denominator. In an orthonormal eigenbasis write $\mathbf{f} = \sum_{j=2}^{M} c_j \mathbf{f}_j$. Then the quotient equals

$$\frac{\sum_{j=2}^{M} (1 - \lambda_j) c_j^2}{\sum_{j=2}^{M} c_j^2} \geq 1 - \lambda_2.$$

Equality holds for $\mathbf{f} = \mathbf{f}_2$. ∎

The next two sections estimate this quotient without diagonalizing $\mathbf{P}$.

3.3 Cheeger's inequality: bottlenecks in the state space

For a finite irreducible reversible chain, define the stationary flow

$$\Lambda(A, A^c) = \sum_{\mathbf{e} \in A} \sum_{\mathbf{e}' \in A^c} \pi(\mathbf{e}) \mathbf{P}(\mathbf{e}, \mathbf{e}')$$

and Cheeger's constant (also called conductance) by

$$\gamma_C = \min_{A : 0 < \pi(A) \leq 1/2} \frac{\Lambda(A, A^c)}{\pi(A)}. \tag{3.17}$$

The ratio is the probability of leaving A in one step when the starting law is $\boldsymbol{\pi}$ conditioned on A. A small ratio identifies a bottleneck: substantial stationary mass can stay on one side of a narrow boundary.

Theorem 3.3.1 — Cheeger's inequality. For a finite irreducible reversible chain,

$$\frac{\gamma_C^2}{2} \le \gamma = 1 - \lambda_2 \le 2\gamma_C. \tag{3.18}$$

Equivalently, $1 - 2\gamma_C \le \lambda_2 \le 1 - \gamma_C^2/2$.

This is a bound for λ_2 in decreasing real-number order. For a lazy chain it also bounds λ_* and hence the TV rate. For background and further examples, see [LP17, DS91].

3.3.1 The easy bound: use an indicator function

Proof of $\gamma \le 2\gamma_C$. For an admissible set A, take $\mathbf{f} = (\mathbf{1}_{\{\mathbf{e}\in A\}})_{\mathbf{e}\in\mathbb{E}}$. Reversibility gives

$$\mathscr{E}(\mathbf{f},\mathbf{f}) = \Lambda(A,A^c), \qquad \mathrm{Var}_{\boldsymbol{\pi}}(f) = \pi(A)\pi(A^c).$$

The variational formula (3.16) and $\pi(A^c) \ge 1/2$ imply

$$\gamma \le \frac{\Lambda(A,A^c)}{\pi(A)\pi(A^c)} \le 2\frac{\Lambda(A,A^c)}{\pi(A)}.$$

Minimise over A. ∎

3.3.2 The lower bound via level sets (optional)

The proof in this subsection is optional for the course. We give the argument with the constants stated in (3.18).

Lemma 3.3.2 — A bound for a function supported on half the stationary mass. If $g \ge 0$, $g \not\equiv 0$, and $\pi\{g > 0\} \le 1/2$, then

$$\mathscr{E}(\mathbf{g},\mathbf{g}) \ge \frac{\gamma_C^2}{2}\|\mathbf{g}\|_{\boldsymbol{\pi}}^2. \tag{3.19}$$

Proof. For $t \ge 0$, put $A_t = \{\mathbf{e} : g(\mathbf{e})^2 > t\}$. The elementary identity $|u - v| = \int_0^\infty |\mathbf{1}_{\{u>t\}} - \mathbf{1}_{\{v>t\}}|\,dt$ for $u, v \ge 0$ gives

$$\begin{aligned} S &:= \sum_{\mathbf{e},\mathbf{e}'} \Lambda(\mathbf{e},\mathbf{e}')|g(\mathbf{e})^2 - g(\mathbf{e}')^2| \\ &= 2\int_0^\infty \Lambda(A_t, A_t^c)\,dt \\ &\ge 2\gamma_C \int_0^\infty \pi(A_t)\,dt = 2\gamma_C\|\mathbf{g}\|_{\boldsymbol{\pi}}^2. \end{aligned}$$

Empty level sets contribute zero. Each other level set has stationary mass at most $1/2$, so we may apply (3.17). On the other hand, Cauchy–Schwarz gives

$$\begin{aligned} S^2 &\le \left[\sum_{\mathbf{e},\mathbf{e}'} \Lambda(\mathbf{e},\mathbf{e}')(g(\mathbf{e}) - g(\mathbf{e}'))^2\right]\left[\sum_{\mathbf{e},\mathbf{e}'} \Lambda(\mathbf{e},\mathbf{e}')(g(\mathbf{e}) + g(\mathbf{e}'))^2\right] \\ &\le 2\mathscr{E}(\mathbf{g},\mathbf{g}) \cdot 4\|\mathbf{g}\|_{\boldsymbol{\pi}}^2. \end{aligned}$$

For the last factor use $(u+v)^2 \le 2(u^2+v^2)$ and the two marginal sums $\sum_{\mathbf{e}'} \Lambda(\mathbf{e},\mathbf{e}') = \pi(\mathbf{e})$ and $\sum_{\mathbf{e}} \Lambda(\mathbf{e},\mathbf{e}') = \pi(\mathbf{e}')$. Combining the two estimates and cancelling the positive norm proves the lemma. ∎

Proof of $\gamma \geq \gamma_C^2/2$. Take a real eigenvector $\boldsymbol{\psi}$ for λ_2, with $\boldsymbol{\pi}$-mean zero. Choose its sign so that $0 < \pi\{\psi > 0\} \leq 1/2$, and put $g(\mathbf{e}) = \max\{\psi(\mathbf{e}), 0\}$. On the set where $g(\mathbf{e}) > 0$,

$$((\mathbf{I}-\mathbf{P})\mathbf{g})(\mathbf{e}) \leq ((\mathbf{I}-\mathbf{P})\boldsymbol{\psi})(\mathbf{e}) = \gamma\psi(\mathbf{e}) = \gamma g(\mathbf{e}),$$

since $g(\mathbf{e}') \geq \psi(\mathbf{e}')$ for all $\mathbf{e}'$. Multiplying by $\pi(\mathbf{e})g(\mathbf{e})$ and summing gives

$$\mathscr{E}(\mathbf{g},\mathbf{g}) \leq \gamma\|\mathbf{g}\|_{\boldsymbol{\pi}}^2. \tag{3.20}$$

Lemma 3.3.2 gives the opposite bound with $\gamma_C^2/2$ in place of γ. Cancel the nonzero squared norm. ■

3.3.3 A TV estimate and the walk on a circle

Exercise 3.5 For a finite irreducible lazy reversible chain, show that for $0 < \varepsilon < 1/2$ and a starting state $\mathbf{e}$,

$$d_{TV}(\boldsymbol{\delta}_{\mathbf{e}}\mathbf{P}^n, \boldsymbol{\pi}) \leq \varepsilon \quad \text{if} \quad n \geq \left\lceil \frac{2}{\gamma_C^2} \log \frac{1}{2\varepsilon\sqrt{\pi(\mathbf{e})}} \right\rceil. \tag{3.21}$$

Use Corollary 3.2.3 and $1-u \leq e^{-u}$. ■

Example 3.5 — A lazy symmetric random walk on a circle. Let $\mathbb{E} = \{0,\dots,n-1\}$, $n \geq 3$, and choose $0 < p \leq 1/4$. Move one step left or right modulo n, each with probability p, and stay put with probability $1-2p$. Then

$$\mathbf{P}(i,i) = 1-2p, \qquad \mathbf{P}(i,i\pm 1) = p, \qquad \pi(i) = 1/n.$$

For a nonempty proper set A, at least two circle edges cross its boundary. A consecutive segment has exactly two, so the smallest boundary flow is $2p/n$. Under $|A| \leq n/2$, the largest allowed set size is $\lfloor n/2 \rfloor$. Therefore

$$\gamma_C = \frac{2p}{\lfloor n/2 \rfloor} = \begin{cases} 4p/n, & n \text{ even}, \\ 4p/(n-1), & n \text{ odd}. \end{cases}$$

A segment achieving this value is $A = \{0,1,\dots,\lfloor n/2 \rfloor - 1\}$. For even n, (3.21) gives the sufficient number of steps

$$\left\lceil \frac{n^2}{8p^2} \left(\frac{1}{2}\log n + \log \frac{1}{2\varepsilon} \right) \right\rceil.$$

For fixed p, ε this is an $O(n^2 \log n)$ upper bound. The SST construction later in this chapter gives an $O(n^2)$ bound for $p = 1/4$ and n a power of two. The extra logarithm here comes from using a general spectral-to-TV estimate.

3.4 Poincaré's inequality and canonical paths

We continue with a finite irreducible reversible chain. The variational formula (3.16) says that a bound

$$\mathrm{Var}_{\boldsymbol{\pi}}(f) \leq \gamma_P \mathscr{E}(\mathbf{f},\mathbf{f})$$

for all functions gives $\gamma \geq 1/\gamma_P$. We obtain such a bound by routing paths between pairs of states.

3.4.1 Congestion of a system of paths

Let $K = \{(\mathbf{e}_i, \mathbf{e}_j) : \mathbf{e}_i \neq \mathbf{e}_j,\ \mathbf{P}(\mathbf{e}_i, \mathbf{e}_j) > 0\}$ be the set of directed edges. For each ordered pair of distinct states $(\mathbf{e}, \mathbf{e}')$ choose a path $\Gamma(\mathbf{e}, \mathbf{e}')$ from $\mathbf{e}$ to $\mathbf{e}'$ using these edges, without repeated vertices. Let $|\Gamma(\mathbf{e}, \mathbf{e}')|$ denote its number of edges. Define

$$\gamma_P = \max_{e \in K} \frac{1}{\Lambda(e)} \sum_{(\mathbf{e},\mathbf{e}'):\, e \in \Gamma(\mathbf{e},\mathbf{e}')} |\Gamma(\mathbf{e}, \mathbf{e}')| \pi(\mathbf{e}) \pi(\mathbf{e}'), \tag{3.22}$$

where $\Lambda(e) = \pi(\mathbf{e}_i)\mathbf{P}(\mathbf{e}_i, \mathbf{e}_j)$ for $e = (\mathbf{e}_i, \mathbf{e}_j)$. The numerator measures how much path traffic passes through an edge; the denominator is the edge's stationary flow. The use of directed edges keeps the factor $1/2$ in the Dirichlet form consistent with the sum over ordered state pairs.

Theorem 3.4.1 — Canonical-path Poincaré bound. For any such system of paths,

$$\mathrm{Var}_{\boldsymbol{\pi}}(f) \leq \gamma_P \mathscr{E}(\mathbf{f}, \mathbf{f}), \qquad \gamma = 1 - \lambda_2 \geq 1/\gamma_P.$$

For a lazy chain, this also gives $\lambda_* \leq 1 - 1/\gamma_P$.

Proof. For $e = (\mathbf{e}_i, \mathbf{e}_j)$ put $\nabla_e f = f(\mathbf{e}_i) - f(\mathbf{e}_j)$. Along a path the differences telescope, so Cauchy–Schwarz gives

$$(f(\mathbf{e}) - f(\mathbf{e}'))^2 = \left(\sum_{e \in \Gamma(\mathbf{e},\mathbf{e}')} \nabla_e f \right)^2 \leq |\Gamma(\mathbf{e}, \mathbf{e}')| \sum_{e \in \Gamma(\mathbf{e},\mathbf{e}')} (\nabla_e f)^2.$$

Use (3.15) and interchange the finite sums:

$$\begin{aligned} \mathrm{Var}_{\boldsymbol{\pi}}(f) &\leq \frac{1}{2} \sum_{e \in K} (\nabla_e f)^2 \sum_{(\mathbf{e},\mathbf{e}'):\, e \in \Gamma(\mathbf{e},\mathbf{e}')} |\Gamma(\mathbf{e}, \mathbf{e}')| \pi(\mathbf{e}) \pi(\mathbf{e}') \\ &\leq \frac{\gamma_P}{2} \sum_{e \in K} \Lambda(e) (\nabla_e f)^2 = \gamma_P \mathscr{E}(\mathbf{f}, \mathbf{f}). \end{aligned}$$

Apply (3.16). For a lazy chain $\lambda_* = \lambda_2$. ∎

Exercise 3.6 For an irreducible lazy reversible chain, a starting state $\mathbf{e}$ and $0 < \varepsilon < 1/2$, show that

$$d_{TV}(\boldsymbol{\delta}_{\mathbf{e}} \mathbf{P}^n, \boldsymbol{\pi}) \leq \varepsilon \quad \text{if} \quad n \geq \left\lceil \gamma_P \log \frac{1}{2\varepsilon \sqrt{\pi(\mathbf{e})}} \right\rceil. \tag{3.23}$$

■

3.4.2 Example: paths on the hypercube

Consider the second hypercube walk from Section 3.1.6. Its state space is $\mathbb{E} = \{0,1\}^d$ and

$$\mathbf{P}(\mathbf{e}, \mathbf{e}') = \begin{cases} 1/(d+1), & \mathbf{e}' = \mathbf{e} \text{ or } \mathbf{e}' \text{ differs in one coordinate,} \\ 0, & \text{otherwise.} \end{cases} \tag{3.24}$$

The stationary law is uniform: $\pi(\mathbf{e}) = 2^{-d}$. Choose the path from $\mathbf{e}$ to $\mathbf{e}'$ which changes differing coordinates from left to right. For example,

$$\begin{aligned} (1,1,0,1,1,1,0,1) &\to (1,0,0,1,1,1,0,1) \\ &\to (1,0,0,0,1,1,0,1) \\ &\to (1,0,0,0,0,1,0,1) \\ &\to (1,0,0,0,0,0,0,1) \to (1,0,0,0,0,0,0,0). \end{aligned}$$

Write $\mathbf{e} = (e_1, \dots, e_d)$ and $\mathbf{e}' = (e'_1, \dots, e'_d)$. Fix a directed edge $e = (\mathbf{a}, \mathbf{b})$ that changes coordinate i. A path using e must satisfy

$$e_i = a_i, \quad e'_i = b_i, \quad (e_{i+1}, \dots, e_d) = (a_{i+1}, \dots, a_d), \quad (e'_1, \dots, e'_{i-1}) = (a_1, \dots, a_{i-1}).$$

Only $e_1, \dots, e_{i-1}$ and $e'_{i+1}, \dots, e'_d$ are free. There are therefore $2^{i-1}2^{d-i} = 2^{d-1}$ ordered pairs whose path uses e. Every path has length at most d, and $\Lambda(e) = 2^{-d}/(d+1)$, so

$$\gamma_P \le 2^d (d+1)\, 2^{d-1} d\, 2^{-2d} = \frac{d(d+1)}{2}. \tag{3.25}$$

Hence $\lambda_2 \le 1 - 2/[d(d+1)]$.

This walk has holding probability $1/(d+1)$ and is not lazy when $d > 1$. To use the bound for TV we must also control its negative eigenvalues. For $A \subseteq \{1, \dots, d\}$, the functions

$$f_A(\mathbf{e}) = (-1)^{\sum_{i \in A} e_i}$$

form an eigenbasis, with eigenvalues $1 - 2|A|/(d+1)$. Thus for $d \ge 2$,

$$\lambda_* = (d-1)/(d+1) = \lambda_2;$$

for $d = 1$, the nontrivial eigenvalue is zero. In this example the canonical-path bound therefore also controls λ_*. Using (3.13) and (3.25), a sufficient number of steps is

$$\left\lceil \frac{d(d+1)}{2} \left(\frac{d}{2} \log 2 + \log \frac{1}{2\varepsilon} \right) \right\rceil.$$

For fixed ε this gives an $O(d^3)$ upper bound. It does not claim that the actual mixing time has order d^3; the coupling from Section 3.1.6 gives a better estimate. The comparison shows how the choice of method affects the bound.

3.5 Non-reversible chains: multiplicative reversiblization

For a non-reversible transition matrix, we can estimate TV convergence using a reversible matrix built from it. Let $\mathbf{P}$ be finite and irreducible, with stationary law $\boldsymbol{\pi}$. Its time reversal $\widetilde{\mathbf{P}}$, defined in Section 3.1.8, is the adjoint of $\mathbf{P}$ in the $\boldsymbol{\pi}$-weighted inner product:

$$\langle \mathbf{f}, \mathbf{P}\mathbf{g} \rangle_{\boldsymbol{\pi}} = \langle \widetilde{\mathbf{P}}\mathbf{f}, \mathbf{g} \rangle_{\boldsymbol{\pi}}.$$

Define the multiplicative reversiblization

$$\mathbf{M} = \mathbf{P}\widetilde{\mathbf{P}}.$$

It is stochastic, has stationary law $\boldsymbol{\pi}$, and is self-adjoint and positive semidefinite in this inner product. Its eigenvalues lie in $[0, 1]$. Write $\lambda_2(\mathbf{M})$ for its second eigenvalue in decreasing order, allowing $\lambda_2(\mathbf{M}) = 1$ if the stationary eigenspace is not simple.

Lemma 3.5.1 For any column of function values $\mathbf{f}$,

$$\langle \mathbf{f}, \mathbf{M}\mathbf{f} \rangle_{\boldsymbol{\pi}} = \|\widetilde{\mathbf{P}}\mathbf{f}\|^2_{\boldsymbol{\pi}}.$$

Proof. Apply the adjoint identity to $\mathbf{P}\widetilde{\mathbf{P}}$. ∎

Theorem 3.5.2 — Mihail's identity. For any real function f,

$$\mathrm{Var}_{\boldsymbol{\pi}}(f) = \mathrm{Var}_{\boldsymbol{\pi}}(\widetilde{\mathbf{P}}\mathbf{f}) + \mathscr{E}_{\mathbf{M}}(\mathbf{f},\mathbf{f}). \tag{3.26}$$

In the first term on the right, the variance is that of the function with values $\widetilde{\mathbf{P}}\mathbf{f}$.

Proof. Subtract the mean, which both matrices preserve. For a mean-zero $\mathbf{f}$, Lemma 3.5.1 gives

$$\mathscr{E}_{\mathbf{M}}(\mathbf{f},\mathbf{f}) = \langle \mathbf{f}, (\mathbf{I}-\mathbf{M})\mathbf{f}\rangle_{\boldsymbol{\pi}} = \|\mathbf{f}\|_{\boldsymbol{\pi}}^2 - \|\widetilde{\mathbf{P}}\mathbf{f}\|_{\boldsymbol{\pi}}^2.$$

These squared norms are the corresponding variances. ∎

For $\boldsymbol{\mu}_n = \boldsymbol{\mu}\mathbf{P}^n$, define the density column and chi-square discrepancy

$$\boldsymbol{\rho}_n = (\mu_n(\mathbf{e})/\pi(\mathbf{e}))_{\mathbf{e}\in\mathbb{E}}, \qquad \chi_n^2 = \sum_{\mathbf{e}} \frac{(\mu_n(\mathbf{e})-\pi(\mathbf{e}))^2}{\pi(\mathbf{e})} = \|\boldsymbol{\rho}_n - \mathbf{1}\|_{\boldsymbol{\pi}}^2.$$

Time reversal gives $\boldsymbol{\rho}_{n+1} = \widetilde{\mathbf{P}}\boldsymbol{\rho}_n$.

Theorem 3.5.3 — A reversiblization bound; cf. Fill (Fil91). For every initial distribution,

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}^n, \boldsymbol{\pi}) \le \frac{1}{2}\chi_n \le \frac{1}{2}\lambda_2(\mathbf{M})^{n/2}\chi_0. \tag{3.27}$$

In particular, for a starting state $\mathbf{e}$, $\chi_0 = \sqrt{1/\pi(\mathbf{e})-1}$.

Proof. The mean-zero variational bound for $\mathbf{M}$ gives

$$\mathscr{E}_{\mathbf{M}}(\boldsymbol{\rho}_n, \boldsymbol{\rho}_n) \ge (1-\lambda_2(\mathbf{M}))\chi_n^2.$$

By (3.26), $\chi_{n+1}^2 \le \lambda_2(\mathbf{M})\chi_n^2$. Iterate this inequality. For the TV estimate apply Cauchy–Schwarz:

$$2d_{TV}(\boldsymbol{\mu}_n, \boldsymbol{\pi}) = \sum_{\mathbf{e}} \sqrt{\pi(\mathbf{e})}\frac{|\mu_n(\mathbf{e})-\pi(\mathbf{e})|}{\sqrt{\pi(\mathbf{e})}} \le \chi_n.$$

For a point mass, direct summation gives $\chi_0^2 = (1-\pi(\mathbf{e}))^2/\pi(\mathbf{e}) + \sum_{\mathbf{e}'\neq\mathbf{e}}\pi(\mathbf{e}') = 1/\pi(\mathbf{e}) - 1$. ∎

Remark 3.5.1 — When the bound contracts. Even if $\mathbf{P}$ is ergodic, $\mathbf{M}$ can be reducible, giving $\lambda_2(\mathbf{M}) = 1$. The bound remains valid but then supplies no decay. We can instead apply the argument to $\mathbf{P}^m$ for a positive power: $\mathbf{M}_m = \mathbf{P}^m\widetilde{\mathbf{P}}^m$ has positive entries and $\lambda_2(\mathbf{M}_m) < 1$. Cheeger's or Poincaré's inequality can bound this reversible matrix. Contractivity covers the remaining steps between blocks.

Remark 3.5.2 — The reversible case. If $\mathbf{P} = \widetilde{\mathbf{P}}$, then $\mathbf{M} = \mathbf{P}^2$ and $\lambda_2(\mathbf{M}) = \lambda_*^2$. Equation (3.27) recovers Corollary 3.2.3, including the role of negative eigenvalues of the original matrix.

3.6 Strong stationary times

A strong stationary time certifies randomisation by a stopping rule on a single chain. It gives a convergence bound through the tail of that stopping time, without estimating eigenvalues.

3.6.1 Motivating example: top-to-random shuffling

In Section 3.1.4 we introduced *Random-to-Top* card shuffling. Roughly speaking, at each step a random card (or position) was chosen and placed on top. In Example 3.2 we introduced *Top-to-Random*, which at one step put top card into a random position.

Both procedures are similar. More exactly, one is a time reversal of the other, thus – since both are random walks on groups – the rate of convergence of one of the procedure is the same as the other one (recall Lemma 3.1.6). Thus, we have already shown a rate of convergence of Top-to-Random using the coupling for Random-to-Top (Example 3.2). We now analyse top-to-random directly by observing the original bottom card. The running time of the resulting algorithm will be a strong stationary time.

Assume n cards are initially (this is X_0) in the following ordering (_ represent possible gaps where top card can be inserted):

$$c_1,_,c_2,_,c_3,_,\dots,_,c_{n-1},_,c_n,_$$

Consider the following random variables $T_1,\dots,T_n$:

Algorithm 3.1

- Observe $X=(X_0,X_1,\dots,X_m)$ till first card gets under c_n (denote this moment by T_1)
- Observe random walk till second card gets under c_n (denote this moment by T_2)

$\vdots$

- Observe random walk till there will be $n-1$ cards under c_n (i.e. card c_n is on the top, denote this moment by T_{n-1})
- Do one step more = T_n
- **Stop**

At time T_2 there are two cards under c_n, assume they are c_i,c_j. There are two possibilities of their relative ordering:

$$\dots,c_n,c_i,c_j \quad \text{or} \quad \dots,c_n,c_j,c_i$$

Assume c_i was under c_n at time T_1. At T_2 card c_j could get under or over c_i with the same probability $1/n$, thus both arrangements (c_i,c_j) and (c_j,c_i) are equally likely (having probability $1/2$).

Now, at T_3 a third card, say c_m gets under c_n, with equal probability it can get to any 3 possible positions, thus all 6 arrangements

$$(c_i,c_j,c_m),(c_i,c_m,c_j),(c_j,c_i,c_m),(c_j,c_m,c_i),(c_m,c_i,c_j),(c_m,c_j,c_i)$$

are equally likely, all having probability $1/6$. By induction, at time T_{n-1} the card c_n is on the top, and all $(n-1)!$ arrangements of the remaining cards are equally likely. Thus, if we put c_n in the random position (next step $T_n=T_{n-1}+1$), all $n!$ arrangements are equally likely. At this moment cards are "perfectly shuffled". The same induction works conditional on the arrival times: conditional on a new card entering the region below c_n, its insertion slot is uniform among the slots in that region. The waiting time for that arrival depends only on the number of slots, not on their relative order. This gives uniformity conditional on T_n, the strong property needed below. Let us compute $\mathbb{E}T_n$. Denote $T_0=0$, we can decompose

$$T=\underbrace{T_n-T_{n-1}}_{X_n}+\underbrace{T_{n-1}-T_{n-2}}_{X_{n-1}}+\dots+\underbrace{T_1-T_0}_{X_1}, \tag{3.28}$$

where $X_i=T_i-T_{i-1}\sim Geo(\frac{i}{n})$, and X_i are independent. It is easy to compute

$$\mathbb{E}T=\sum_{i=1}^{n}\mathbb{E}X_i=\sum_{i=1}^{n}\frac{n}{i}=nH_n\sim n\log n.$$

The next subsection formalizes this stopping rule and uses it to bound $d_{TV}(\boldsymbol{\mu}\mathbf{P}^k,\boldsymbol{\pi})$.

3.6.2 Definition, separation and the tail bound

A stopping time T takes values in $\{0,1,\dots\}\cup\{\infty\}$ and the question whether $T \le n$ can be answered from the history through time n. A randomized stopping time may also use auxiliary independent random coins. For example, after observing $X_0,\dots,X_n$, an algorithm can stop with probability $f_n(X_0,\dots,X_n)$, using a fresh coin independent of future chain transitions. If these probabilities are only zero or one, the rule is non-randomized.

Algorithm 3.2 — A randomized stopping rule. After each observed step, compute the stopping probability from the current history, flip an independent coin with that probability, and stop on success. Otherwise observe the next step and repeat.

Definition 3.6.1 — Strong stationary time. A randomized stopping time T for a chain with initial law $\boldsymbol{\mu}$ and stationary law $\boldsymbol{\pi}$ is a strong stationary time (SST) if, for every $n \ge 0$ and $\mathbf{e} \in \mathbb{E}$,

$$\mathbb{P}_\mu(X_n = \mathbf{e}, T = n) = \pi(\mathbf{e})\mathbb{P}_\mu(T = n). \tag{3.29}$$

For an SST used to sample from $\boldsymbol{\pi}$ we also require $T < \infty$ almost surely.

Whenever $\mathbb{P}(T=n) > 0$, this says $\mathbb{P}(X_n = \mathbf{e} \mid T = n) = \pi(\mathbf{e})$. Thus the state at stopping is stationary and independent of the stopping time. Merely having $X_T \sim \boldsymbol{\pi}$ is not sufficient for the strong property.

For probability distributions $\boldsymbol{\mu}, \boldsymbol{\nu}$ with $\nu(\mathbf{e}) > 0$ for all $\mathbf{e}$, define separation by

$$sep(\boldsymbol{\mu}, \boldsymbol{\nu}) = \max_{\mathbf{e}\in\mathbb{E}} \left(1 - \frac{\mu(\mathbf{e})}{\nu(\mathbf{e})}\right).$$

It lies in $[0,1]$ but is not symmetric, so it is not a metric. For distance from stationarity, the positive stationary distribution makes all the ratios well-defined.

Exercise 3.7

$$d_{TV}(\boldsymbol{\mu}, \boldsymbol{\nu}) \le sep(\boldsymbol{\mu}, \boldsymbol{\nu}). \tag{3.30}$$

If $s = sep(\boldsymbol{\mu}, \boldsymbol{\nu})$, use $\mu(\mathbf{e}) \ge (1-s)\nu(\mathbf{e})$ and write $\boldsymbol{\mu} = (1-s)\boldsymbol{\nu} + s\mathbf{r}$ for a probability distribution $\mathbf{r}$ when $s > 0$. ∎

Lemma 3.6.1 — SST tail bound; Aldous and Diaconis (AD87). For any strong stationary time,

$$sep(\boldsymbol{\mu}\mathbf{P}^n, \boldsymbol{\pi}) \le \mathbb{P}(T > n). \tag{3.31}$$

Proof. For $j \le n$, the stopping-time property and the Markov property give

$$\mathbb{P}(X_n = \mathbf{e} \mid T = j, X_j = \mathbf{e}') = \mathbf{P}^{n-j}(\mathbf{e}', \mathbf{e}) \tag{3.32}$$

whenever the conditioning event has positive probability. The auxiliary stopping coins reveal no

information about future transitions. Use (3.29) and stationarity to obtain

$$\begin{aligned}\mathbb{P}(X_n = \mathbf{e}) &\geq \sum_{j=0}^{n} \mathbb{P}(X_n = \mathbf{e}, T = j) \\ &= \sum_{j=0}^{n} \sum_{\mathbf{e}'} \mathbf{P}^{n-j}(\mathbf{e}', \mathbf{e}) \pi(\mathbf{e}') \mathbb{P}(T = j) \\ &= \pi(\mathbf{e}) \mathbb{P}(T \leq n).\end{aligned}$$

Divide by $\pi(\mathbf{e})$ and maximise over $\mathbf{e}$. ■

Combining this with (3.30) gives

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}^n, \boldsymbol{\pi}) \leq sep(\boldsymbol{\mu}\mathbf{P}^n, \boldsymbol{\pi}) \leq \mathbb{P}(T > n). \tag{3.33}$$

Example 3.6 — Top-to-random: mean, variance and a tail bound. The running time of Algorithm 3.1 is a strong stationary time for Top-to-Random card shuffling. (the intuitive explanation presented after this algorithm was actually the proof). We showed that

$$T = \sum_{i=1}^{n} X_i, \quad X_i \sim Geo(i/n),$$

and computed that $\mathbb{E}T = nH_n \sim n \log n$. The variance of a geometric random variable $\mathrm{Var}(X_i) = (1 - i/n)/(i/n)^2$, thus

$$\begin{aligned}\mathrm{Var}\, T = \sum_{i=1}^{n} \mathrm{Var}\, X_i = \sum_{i=1}^{n} \frac{1 - \frac{i}{n}}{(\frac{i}{n})^2} = \sum_{i=1}^{n} \left(1 - \frac{i}{n}\right) \left(\frac{n}{i}\right)^2 = \\ = n^2 \sum_{i=1}^{n} \frac{1}{i^2} - n \sum_{i=1}^{n} \frac{1}{i} \approx \frac{\pi^2}{6} n^2 - n \log n \approx n^2\end{aligned}$$

For $c > 0$, Chebyshev's inequality gives

$$\mathbb{P}(T > \mathbb{E}T + cn) \leq \frac{\mathrm{Var}\, T}{c^2 n^2} \leq \frac{\pi^2}{6c^2}.$$

The coupon-collector description gives the sharper bound

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}^{\lceil n \log n + cn \rceil}, \boldsymbol{\pi}) \leq sep(\boldsymbol{\mu}\mathbf{P}^{\lceil n \log n + cn \rceil}, \boldsymbol{\pi}) \leq e^{-c}.$$

Exercise 3.8 Find a strong stationary time T for a symmetric random walk on d-dimensional cube, ver1, presented in Section 3.1.5. ■

Exercise 3.9 — *. Find a strong stationary time T for a symmetric random walk on d-dimensional cube, ver2, presented in Section 3.1.6. ■

3.6.3 A walk on perfect matchings

Definition 3.6.2 — Perfect matching. A perfect matching of a graph is a set of disjoint edges covering all vertices. For a complete graph on $2n$ vertices, it is a partition of the vertices into n unordered pairs.

The three matchings on four vertices are shown in Figure 3.1.

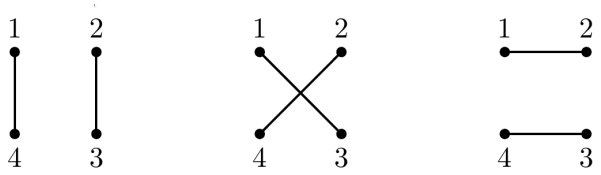


Figure 3.1: Perfect matchings of the complete graph on four vertices.

Exercise 3.10 Show that the number of perfect matchings on $2n$ labelled vertices is $(2n-1)!! = (2n)!/(2^n n!)$. ∎

The state space $\mathbb{E}$ consists of these matchings. From a current matching $\mathbf{s}$, choose two of its pairs L, R independently and uniformly. If $L = R$, keep the matching. Otherwise, for $L = \{a,b\}$ and $R = \{c,d\}$, choose with equal probabilities between the two replacements

$$\{\{a,c\},\{b,d\}\}, \qquad \{\{a,d\},\{b,c\}\}.$$

The remaining pairs stay unchanged. See Figure 3.2.

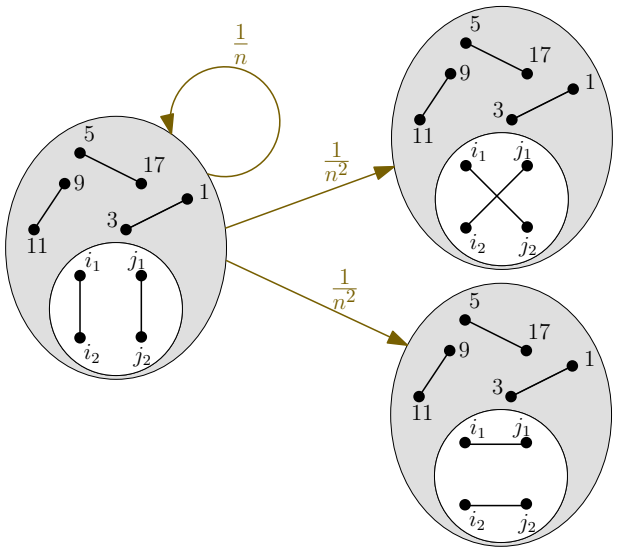


Figure 3.2: One step of the walk on matchings.

The transition matrix satisfies

$$\mathbf{P}(\mathbf{s},\mathbf{s}') = \begin{cases} 1/n, & \mathbf{s}' = \mathbf{s}, \\ 1/n^2, & \mathbf{s}' \text{ is obtained by one of the two-pair replacements}, \\ 0, & \text{otherwise}. \end{cases} \tag{3.34}$$

For each target replacement, the two orders of selecting the pairs and the fair coin give $2 \cdot n^{-2} \cdot (1/2) = n^{-2}$. There are $2\binom{n}{2}$ distinct neighbours, so the row sum is $1/n + 2\binom{n}{2}/n^2 = 1$. The matrix is symmetric; hence the stationary law is uniform. The walk is irreducible, since such replacements can create the pairs of any target matching one at a time, and its holding probability makes it aperiodic.

Algorithm 3.3 — Marking pairs. Start with no marked vertices. At each step choose L, R and perform the walk's update, with the following marking rule:

- If $L = R$ is unmarked, mark its two vertices.
- If exactly one of L, R is marked, mark all four vertices in the two resulting pairs. This increases the number of marked pairs by one.
- Otherwise keep the marked vertex set unchanged.

Stop when all vertices are marked.

All current pairs have either two marked vertices or two unmarked vertices. Let T_i be the time when i pairs first become marked, with $T_0 = 0$, and let $T = T_n$.

Lemma 3.6.2 The running time T of Algorithm 3.3 is a strong stationary time.

Proof. We maintain the following property: conditional on the marked vertex set, the unmarked matching and the marking times, the matching of marked vertices is uniform. This is immediate with zero or one marked pair. A step involving only marked pairs uses a symmetric transition on their matchings and preserves uniformity. A step involving only unmarked pairs, without a new mark, does not change the marked matching.

Consider a marking step when r pairs are marked, and fix the unmarked pair $\{a, b\}$ which joins them. Selecting that pair twice gives the extension in which a, b stay paired, with probability $1/n^2$. For each of the r marked pairs, selecting it together with $\{a, b\}$ and using either replacement gives two further extensions, each with probability $1/n^2$. Thus the $2r+1$ extensions are equally likely.

Each target matching of the enlarged marked vertex set has one predecessor in this description. If a, b are paired in the target, remove that pair. Otherwise, let u, v be their target partners and replace the two target pairs by $\{u, v\}$ to recover the predecessor on the old marked vertices. Uniform predecessors and equal extension probabilities therefore give a uniform matching on the enlarged set.

The probability of a marking step depends only on r, not on the marked matching, so conditioning on the waiting times preserves this property. At T, the marked set is the whole vertex set. Conditional on $T = k$, the matching is consequently uniform on $\mathbb{E}$, which is (3.29). ■

Exercise 3.11 Show that $T = \sum_{i=1}^{n} (T_i - T_{i-1})$ has independent geometric summands. When $r = i - 1$ pairs are marked, count the marking moves to obtain

$$p_i = \frac{(n-r)(2r+1)}{n^2}.$$

Compute $\mathbb{E}T = \sum_i p_i^{-1}$ and $\operatorname{Var} T = \sum_i (1-p_i)/p_i^2$. Show that these have orders $n \log n$ and n^2, respectively, as $n \to \infty$. For $c > 0$ deduce

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}^k, \boldsymbol{\pi}) \le c^{-2} \quad \text{if} \quad k \ge \left\lceil \mathbb{E}T + c\sqrt{\operatorname{Var} T} \right\rceil.$$

Thus the construction gives an $O_\varepsilon(n \log n)$ mixing-time upper bound. ■

3.6.4 Example: Random walk on $\mathbb{Z}_4$

Random walk starts at 0 ($X_0 = 0$), $\mathbb{E} = \{0, 1, 2, 3\}$. Later the chain stays at the same state with probability $\frac{1}{2}$ or goes left or right with probability $\frac{1}{4}$. This can be written as:

$$X_k = X_{k-1} + \theta_k \bmod 4,$$

where $\theta_1, \theta_2, \ldots$ are iid random variables with distribution: $\mathbb{P}(\theta_i = 1) = \mathbb{P}(\theta_i = -1) = \frac{1}{4}, \mathbb{P}(\theta_i = 0) = \frac{1}{2}$.

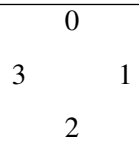


Figure 3.3: $\mathbb{Z}_4$

Here is an algorithm which running time is Strong Uniform Time for above random walk:

Algorithm 3.4 — for Z_4.

- Observe $X = (X_0, X_1, \ldots, X_m)$ till first time when $X_m \in \{1,3\}$
- Do one step more
- **Stop**

Of course the running time of above algorithm is a stopping time. To prove that it is Strong Uniform Time, let us assume that $T = k$. Because of symmetry in step $k-1$ the random walk was either in 1 or in 3 with the same probability, i.e.:

$$\mathbb{P}(X_{k-1} = 1 | T = k) = \mathbb{P}(X_{k-1} = 3 | T = k) = \tfrac{1}{2},$$

Thus for any $i = 0,1,2,3$ we have

$$\mathbb{P}(X_k = i | T = k) = \frac{1}{2}\mathbb{P}(X_{k-1} = i | T = k)$$

$$+\frac{1}{4}[\mathbb{P}(X_{k-1} = i+1 | T = k) + \mathbb{P}(X_{k-1} = i-1 | T = k)] = \tfrac{1}{4} = \tfrac{1}{|\mathbb{E}|}.$$

In order to have a bound on $d_{TV}(\boldsymbol{\mu}\mathbf{P}^n, \boldsymbol{\pi})$ and to use lemma 3.6.1 we have to calculate:

$$\mathbb{P}(T > k) = \sum_{i=k+1}^{\infty} \left(\frac{1}{2}\right)^{i-2} \cdot \frac{1}{2} = 2\sum_{k+1}^{\infty} \left(\frac{1}{2}\right)^{i} = 2 \cdot \frac{\left(\frac{1}{2}\right)^{k+1}}{1-\frac{1}{2}} = 2 \cdot \left(\frac{1}{2}\right)^{k}.$$

Thus:

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}^k, \boldsymbol{\pi}) \leq sep(\boldsymbol{\mu}\mathbf{P}^k, \boldsymbol{\pi}) \leq 2 \cdot \left(\frac{1}{2}\right)^{k}.$$

3.6.5 Example: Random walk on $\mathbb{Z}_n$

The example is similar to the example of Diaconis and Fill [DF90].

Assume $n = 2^m$ with $m \geq 2$. For $n = 16$ it is a random walk on n-points circle:
Similarly as in previous example we can write:

$$X_k = X_{k-1} + \theta_k \bmod n,$$

where $\theta_1, \theta_2, \ldots$ are iid with distribution: $\mathbb{P}(\theta_i = 1) = \mathbb{P}(\theta_i = -1) = \frac{1}{4}, \mathbb{P}(\theta_i = 0) = \frac{1}{2}$.

Here is an algorithm which running time is Strong Uniform Time for above walk:

Algorithm 3.5 — for $\mathbb{Z}_n$.

- Observe $X = (X_0, X_1, \ldots, X_m)$ till first time (T_1) when $X_m \in \{2^{m-2}, 3 \cdot 2^{m-2}\}$
- Observe random walk till first time after T_1 when it gets to state which is in distance of 2^{m-3} from X_{T_1} (call this moment T_2)

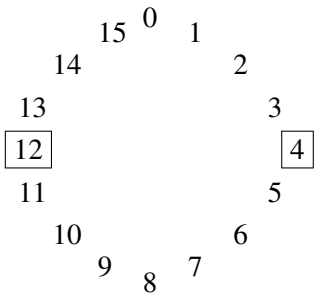


Figure 3.4: $\mathbb{Z}_{16}$

- Observe random walk till first time after T_2 when it gets to state which is in distance of 2^{m-4} from X_{T_2} (call this moment T_3)

$\vdots$

- Observe random walk till first time after T_{m-2} when it gets to state which is in distance of 1 from $X_{T_{m-2}}$ (call this moment T_{m-1} - at this time random walk is in state which is an odd number)
- Observe one step more ($T = T_m$)
- **Stop**

Of course running time of this algorithm is a stopping time. Assume that $T_1 = k_1$, then, because of symmetry, X_{k_1} is in one of the states from $\{2^{m-2}, 3\cdot 2^{m-2}\}$ with equal probability (all paths $i_1, i_2, \ldots, i_{k_1}$ such that at time $k-1$ are first time in state 2^{m-2} or $3\cdot 2^{m-2}$ are equally likely). The same is at moment T_2, and because of symmetry X_{T_2} has uniform distribution on $\{2^{m-2}+2^{m-3}, 2^{m-2}-2^{m-3}, 3\cdot 2^{m-2}+2^{m-3}, 3\cdot 2^{m-2}-2^{m-3}\}$.

So, because of symmetry, at time T_{m-1} the distribution of $X_{T_{m-1}}$ is uniform on all possible odd numbers, i.e.

$$\mathbb{P}(X_k = 2i+1 | T_{m-1} = k) = \frac{1}{2^{m-1}}.$$

To calculate $\mathbb{P}(X_k = i | T = k)$ consider separately cases when i is odd and even. If i is odd, then the step earlier it had to be in the same state and had to stay in it. If i is even, then a step earlier it had to be in some neighbouring odd states and in one step it had to get to i. So we can calculate:

$$\mathbb{P}(X_k = 2i+1 | T = k) = \frac{1}{2}\mathbb{P}(X_{k-1} = 2i+1 | T_{m-1} = k-1) = \frac{1}{2}\cdot\frac{1}{2^{m-1}} = \frac{1}{2^m}.$$

$$\mathbb{P}(X_k = 2i | T = k) = \frac{1}{4}\mathbb{P}(X_{k-1} = 2i-1 | T_{m-1} = k-1) + \frac{1}{4}\mathbb{P}(X_{k-1} = 2i+1 | T_{m-1} = k-1)$$

$$= \frac{1}{4}\cdot\frac{1}{2^{m-1}} + \frac{1}{4}\cdot\frac{1}{2^{m-1}} = \frac{1}{2^m}.$$

Thus T is a Strong Uniform Time. To estimate $\mathbb{P}(T > k)$ we will use Markov's inequality (for $Y \geq 0$, $\mathbb{P}(Y > k) \leq \frac{EY}{k}$) and the following lemma:

Lemma 3.6.3 — Feller (Fel68). For random walk on integers starting from 0 and moving $+1$ or -1 with probability $\frac{\theta}{2}$, and staying in the same state with probability $1-\theta$, expected time of getting to $\pm b$ is $\frac{b^2}{\theta}$.

□

In random walk on $\mathbb{Z}_n$ we first have to get to state in distance $\pm 2^{m-2}$, later to state in distance $\pm 2^{m-3}$ etc., last time to state in distance 1 and to make one step more. Here $\theta = 1/2$, so the expected waiting time for a displacement of 2^r is $2 \cdot 4^r$. Summing the stages and the final extra step gives

$$\mathbb{E}T = 2\sum_{r=0}^{m-2} 4^r + 1 = \frac{n^2}{6} + \frac{1}{3}.$$

For $k = cn^2$, $c \geq 1$ an integer, Markov's inequality gives

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}^k, \boldsymbol{\pi}) \leq sep(\boldsymbol{\mu}\mathbf{P}^k, \boldsymbol{\pi}) \leq \mathbb{P}(T > k) \leq \frac{\mathbb{E}T}{k} = \frac{1}{c}\left(\frac{1}{6} + \frac{1}{3n^2}\right) \leq \frac{1}{5c},$$

where the last inequality uses $n \geq 4$. The construction starts from zero; translating it gives an SST with the same tail from any starting state. Thus it gives an $O_\varepsilon(n^2)$ upper bound on the worst-case mixing time for each fixed error tolerance.

3.6.6 Riffle shuffle

The Riffle Shuffle is one of the most popular for shuffling the deck of cards. Roughly speaking (it will be made clear below) we split the deck into two piles (randomly, but most probably the two piles are equal). Then, the cards are riffled: the probability that a next dropped card will be from a given hand, is proportional to number of cards in this hand.

Consider the following 3 models for card shuffling (each one describes one step)

M1 Let K be a Binomial$(n, 1/2)$ random variable. Assume $K = k$. We split a deck into two piles. First pile consists of top k cards and the remaining cards are the second pile (we do not change the ordering of cards). Now we will "mix" the cards in the following way: We choose $\binom{n}{k}$ places uniformly at random, where we place first pile of k cards (keeping their relative ordering) and in the remaining slots we place the pile of $n-k$ cards (also keeping their relative ordering).

M2 Again, let K be a Binomial$(n, 1/2)$ random variable. Assume $K = k$, and we split a decks into two piles as in M1. Next, we take one pile to left hand, and the other one to right hand. We drop card one after another, at each time choosing a hand with probability proportional to a number of cards within this hand. Assume we have L cards in left hand and R cards in right hand. Then, the first dropped card will be the one from left hand with probability $L/(L+R)$, and it will be from right hand with probability $R/(L+R)$. We keep on continuing until all the cards are dropped.

M3 We describe a *time reversed* procedure. Mark each card with a random bit 0 or 1 independently (i.e., each card is marked with 0 or 1 with probability 1/2). Next, we take out all the cards with assigned bit 0 and place them on the top of the deck, *keeping their relative order*.

Lemma 3.6.4 Models M1 and M2 have the same transition matrix. Model M3 has its time-reversed transition matrix, rather than the same transition matrix.

Exercise 3.12 Prove Lemma 3.6.4. ∎

For the inverse riffle shuffle M3, assign a fresh independent bit to each card at each step. After k steps, each card has a k-bit label; stable sorting orders these labels with the newest bit given highest priority. Stop when all labels are distinct. The event of distinctness does not change under

a permutation of card names, so conditional on the stopping time each of the $n!$ relative orders is equally likely. This is an SST for M3. A union bound over pairs of cards gives

$$\mathbb{P}(T > k) \leq \binom{n}{2} 2^{-k}.$$

Hence $d_{TV}(\boldsymbol{\delta}_{id}\mathbf{P}_{M3}^{k}, \boldsymbol{\pi}) \leq \binom{n}{2}2^{-k}$. Lemma 3.1.6 transfers this TV bound to the ordinary riffle shuffle. It is a simple sufficient bound; the sharper seven-shuffle story in Section 1.2 uses a more detailed analysis.

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Chapter 4

tmp

4.1 A two-state Markov chain

This short example illustrates the notation and the typography used in these notes.

Definition 4.1.1 — Transition matrix. A transition matrix on a finite state space $\mathbb{E} = \{\mathbf{e}_1, \dots, \mathbf{e}_M\}$ is a matrix $\mathbf{P} = (\mathbf{P}(\mathbf{e}_i, \mathbf{e}_j))_{i,j=1}^{M}$ satisfying

$$\mathbf{P}(\mathbf{e}, \mathbf{e}') \geq 0, \qquad \sum_{\mathbf{e}' \in \mathbb{E}} \mathbf{P}(\mathbf{e}, \mathbf{e}') = 1 \quad \text{for every } \mathbf{e} \in \mathbb{E}. \tag{4.1}$$

For a Markov chain $(X_n)_{n \geq 0}$ with transition matrix $\mathbf{P}$, $\mathbf{P}(\mathbf{e}, \mathbf{e}') = \mathbb{P}(X_{n+1} = \mathbf{e}' \mid X_n = \mathbf{e})$ whenever the conditioning event has positive probability.

Example 4.1 — Two states. Let $\mathbb{E} = \{0, 1\}$ and $0 < a, b < 1$. The matrix

$$\mathbf{P} = \begin{pmatrix} 1 - a & a \\ b & 1 - b \end{pmatrix} \tag{4.2}$$

satisfies (4.1), so it is a transition matrix in the sense of Definition 4.1.1. Starting from state 0, the chain moves to state 1 with probability a and stays at 0 with probability $1 - a$.

Theorem 4.1.1 — Stationary distribution. The chain in Example 4.1 has the unique stationary distribution

$$\boldsymbol{\pi} = (\pi_0, \pi_1) = \left(\frac{b}{a + b}, \frac{a}{a + b} \right). \tag{4.3}$$

In particular, $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$ and $\pi_0 + \pi_1 = 1$.

Proof. Using the matrix in (4.2), the stationarity equations reduce to

$$\pi_0 a = \pi_1 b, \tag{4.4}$$

$$\pi_0 + \pi_1 = 1. \tag{4.5}$$

Since $a + b > 0$, solving (4.4)–(4.5) yields (4.3) and proves uniqueness. ∎

Remark 4.1.1 For $a = 1/4$ and $b = 1/2$, Theorem 4.1.1 gives $\boldsymbol{\pi} = (2/3, 1/3)$. The links to Section 4.1, Definition 4.1.1, and equation (4.3) are clickable.

4.2 Convergence to equilibrium

The stationary distribution describes equilibrium. To see how the chain approaches it, we can follow the probability of being in state 1 at each time.

Definition 4.2.1 — Total variation distance. For probability distributions $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ on a finite state space $\mathbb{E}$, define

$$\|\boldsymbol{\mu}-\boldsymbol{\nu}\|_{\mathrm{TV}}=\frac{1}{2}\sum_{\mathbf{e}\in\mathbb{E}}|\mu(\mathbf{e})-\nu(\mathbf{e})|. \tag{4.6}$$

This distance lies in $[0,1]$ and is zero exactly when $\boldsymbol{\mu}=\boldsymbol{\nu}$.

Theorem 4.2.1 — An explicit convergence rate. For the chain of Example 4.1, let $q_n=\mathbb{P}(X_n=1)$. Then, with $\lambda=1-a-b$,

$$q_n=\frac{a}{a+b}+\lambda^n\left(q_0-\frac{a}{a+b}\right),\qquad n\geq 0. \tag{4.7}$$

Consequently, $q_n\to\pi_1$, where $\boldsymbol{\pi}$ is given by (4.3).

Proof. Conditioning on X_n gives

$$\begin{aligned}q_{n+1}&=(1-q_n)a+q_n(1-b)\\&=a+(1-a-b)q_n.\end{aligned}$$

Since $\pi_1=a+\lambda\pi_1$, subtraction yields $q_{n+1}-\pi_1=\lambda(q_n-\pi_1)$. Iteration proves (4.7). The assumptions $0<a,b<1$ imply $|\lambda|<1$. ■

Example 4.2 — A numerical rate. Take $a=1/4$, $b=1/2$, and $X_0=0$. By Theorem 4.2.1,

$$q_n=\frac{1}{3}\left(1-4^{-n}\right).$$

On two states, (4.6) reduces to $|q_n-\pi_1|$. Thus the distance to stationarity is $4^{-n}/3$: after four steps it equals $1/768$.

Remark 4.2.1 When $a+b>1$, the factor λ is negative. The probabilities oscillate around their stationary values, while the magnitude of the error still decreases geometrically.

4.3 Random walks and reversibility

A second family of examples comes from moving randomly between neighbouring vertices of a graph. Here the stationary distribution can often be found without solving a linear system.

Definition 4.3.1 — Detailed balance. A transition matrix $\mathbf{P}$ and a probability distribution $\boldsymbol{\pi}$ satisfy detailed balance if

$$\pi(\mathbf{e})\mathbf{P}(\mathbf{e},\mathbf{e}') = \pi(\mathbf{e}')\mathbf{P}(\mathbf{e}',\mathbf{e}), \qquad \mathbf{e},\mathbf{e}' \in \mathbb{E}. \tag{4.8}$$

A chain with such a distribution is called reversible with respect to $\boldsymbol{\pi}$.

Proposition 4.3.1 Every probability distribution satisfying (4.8) is stationary.

Proof. For every $\mathbf{e}' \in \mathbb{E}$, detailed balance and (4.1) give

$$(\boldsymbol{\pi}\mathbf{P})(\mathbf{e}') = \sum_{\mathbf{e}\in\mathbb{E}} \pi(\mathbf{e})\mathbf{P}(\mathbf{e},\mathbf{e}') = \pi(\mathbf{e}') \sum_{\mathbf{e}\in\mathbb{E}} \mathbf{P}(\mathbf{e}',\mathbf{e}) = \pi(\mathbf{e}').$$

■

Example 4.3 — A walk on a path. On the path with vertices $0,1,2$, move to a uniformly chosen neighbour. Then

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \boldsymbol{\pi} = \left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right). \tag{4.9}$$

For the edge $\{0,1\}$, both sides of (4.8) equal $1/4$; the same holds for $\{1,2\}$. Proposition 4.3.1 therefore verifies stationarity. The central vertex has twice the stationary mass of either endpoint because it has twice as many neighbours.

Remark 4.3.1 Stationarity alone does not guarantee convergence. In Example 4.3, a chain started at 0 can visit 1 only at odd times, and 0 or 2 only at even times. This alternating behaviour prevents convergence to (4.9).

4.4 Making a walk lazy

One way to remove the alternating behaviour of Section 4.3 is to allow the chain to stay at its current state.

Definition 4.4.1 — Lazy transition matrix. Given a transition matrix $\mathbf{P}$, its lazy version is

$$\mathbf{P}_{\mathrm{L}} = \frac{1}{2}(\mathbf{I} + \mathbf{P}), \tag{4.10}$$

where $\mathbf{I}$ is the identity matrix. At each step, toss a fair coin: stay put on one outcome and take one step according to $\mathbf{P}$ on the other.

Corollary 4.4.1 If $\boldsymbol{\pi}$ is stationary for $\mathbf{P}$, it is also stationary for $\mathbf{P}_{\mathrm{L}}$.

Proof. By (4.10), $\boldsymbol{\pi}\mathbf{P}_{\mathrm{L}} = (\boldsymbol{\pi} + \boldsymbol{\pi}\mathbf{P})/2 = \boldsymbol{\pi}$. ∎

Example 4.4 — The lazy path. For the matrix in (4.9), the construction gives

$$\mathbf{P}_{\mathrm{L}} = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 1/2 & 1/2 \end{pmatrix}.$$

Corollary 4.4.1 preserves $\boldsymbol{\pi} = (1/4, 1/2, 1/4)$. Starting at 0, the first two distributions are

$$\boldsymbol{\mu}_1 = \left(\frac{1}{2}, \frac{1}{2}, 0\right), \tag{4.11}$$

$$\boldsymbol{\mu}_2 = \left(\frac{3}{8}, \frac{1}{2}, \frac{1}{8}\right). \tag{4.12}$$

Using Definition 4.2.1, we obtain $\|\boldsymbol{\mu}_1 - \boldsymbol{\pi}\|_{\mathrm{TV}} = 1/4$ and $\|\boldsymbol{\mu}_2 - \boldsymbol{\pi}\|_{\mathrm{TV}} = 1/8$.

Exercise 4.1 For the lazy path started at 0, verify that

$$\boldsymbol{\mu}_n = \left(\frac{1}{4} + 2^{-n-1}, \frac{1}{2}, \frac{1}{4} - 2^{-n-1}\right), \qquad n \geq 1.$$

Deduce the total variation distance to $\boldsymbol{\pi}$ and compare your answer with (4.11) and (4.12). ∎

Remark 4.4.1 These examples provide a sample of definitions, proofs, exercises, matrices, and references. For the first numerical example, return to Example 4.2 on page 58.